MLForge
Data-to-Insight Automation Tool

MAJOR PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report "**MLForge: Data-to-Insight Automation Tool**" is the Bonafide work of **"Piyush Jaloniya(24MCI10042)"** who carried out the project work under my/our supervision.

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ABSTRACT

The amount of data produced in every industry is increasing rapidly, making it important to have tools that help people analyze this information and make decisions. However, working with machine learning in a traditional way can be difficult as it often requires good programming skills, a clear understanding of algorithms, and a lot of time for tasks such as data cleaning and model building. To address these challenges, this project introduces a GUI-based tool to facilitate dataset analysis, model training and evaluation using machine learning techniques.

The tool provides a comprehensive platform where users can upload datasets in multiple formats, perform data preprocessing such as cleaning, handling missing values and feature selection, and perform exploratory data analysis through interactive visualizations. The integrated system allows users to choose from a variety of supervised and unsupervised machine learning algorithms, train models and compare their performance using metrics such as accuracy, precision, recall, F1 scores and visual plots.

The graphical interface eliminates the need for extensive coding knowledge, making machine learning accessible to students, researchers, and professionals who want to quickly extract actionable insights from data. In addition, the platform supports iterative experimentation by enabling users to change parameters, apply feature techniques, and seamlessly retrain models.

By combining simplicity, interactivity and flexibility, this project serves as an educational and practical tool for understanding and applying machine learning concepts. This kind of tool doesn't just save time—it also encourages people to make decisions based on data in real situations. It shows how well-designed software can make advanced analytics accessible to everyone, helping remove the barrier between complex technical knowledge and practical insights.

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CHAPTER1: INTRODUCTION

1.1 Background

These days, virtually everything we do generates data, whether it's using sensor-rich devices, social media, shopping online, or running a business. As a result, people, businesses and organizations generate huge amounts of data every day. While this wave of data brings many benefits, it also has its drawbacks. In areas such as healthcare, finance, marketing and education, it is important to derive meaningful insights from all this information so that we can solve real-world problems, make better decisions and predict what may happen in the future.

Machine learning has become an important tool for analyzing large amounts of data, allowing systems to learn from past information and make predictions or decisions without being explicitly programmed. However, traditional machine learning workflows can be complex, involving tasks such as data cleaning, feature development, model selection, hyperparameter tuning, and performance evaluation. These steps require significant technical knowledge and time, which can be challenging for beginners, students, and non-technical users.

GUI-based tools have emerged as a solution to these challenges. They provide an interactive, visual approach to performing complex machine learning tasks without extensive coding. By bringing together data analysis, visualization and model building on one platform, these tools make machine learning more accessible and help users efficiently extract meaningful insights from their data.

1.2 Problem Statement

Despite the rapid adoption of machine learning, several challenges remain:

Complexity of ML workflows: Non-technical users struggle with coding, library dependencies, and algorithm selection.

Time-consuming tasks: Preprocessing data, training multiple models, and evaluating performance require significant effort.

Lack of visualization: Understanding patterns and trends in data often demands advanced plotting and visualization skills.

Difficult model comparison: Manually comparing multiple ML models to identify the best-performing one is cumbersome.

This project aims to overcome these issues by developing a GUI-based tool that simplifies the entire machine learning pipeline. Users can upload datasets, clean and preprocess data, explore patterns through visualizations, train and evaluate multiple models, and compare their performance—all within a single interactive interface.

1.3 Objectives

The main goals of the project are:

Design and develop a user-friendly GUI tool that integrates data analysis, visualization and machine learning.

Provides easy-to-use data preprocessing features such as handling missing values, coding categorical data, and feature scaling.

Provides interactive visualizations including histograms, scatter plots, box plots, and correlation matrices to understand data set patterns.

Use of a variety of machine learning algorithms (both classification and regression) for model training and prediction.

Allows iterative experimentation, including parameter tuning and retraining models for improved performance.

Making machine learning accessible to beginners, students, researchers and professionals, promoting practical learning and real-world application.

1.4 Scope

The scope of this project includes:

Support for structured datasets: Handles formats such as CSV and Excel.

Machine learning integration: Focuses primarily on supervised learning (classification and regression), with the option to extend to unsupervised learning.

Interactive exploratory data analysis (EDA): Provides visualization through plots, graphs and statistical summaries.

Parameter tuning and experimentation: Supports iterative tuning to improve model performance.

Model comparison: Makes it easy to evaluate and visualize multiple models to identify the model with the best results.

Practical applications: Used in education, research and business analysis.

Future enhancements: This may include advanced ML techniques, real-time data processing, cloud integration, and deployment as web or mobile applications.

1.5 Tools & Technologies Used

Python: Core programming language due to its simplicity, rich libraries, and strong ML ecosystem.

Streamlit: Provides a simple yet powerful framework for building interactive GUIs with minimal coding.

Pandas & NumPy: Used for data manipulation, numerical operations, and handling large datasets efficiently.

Matplotlib & Seaborn: Visualization libraries for creating informative charts and plots for EDA.

Scikit-learn: Implements various ML algorithms, preprocessing techniques, and model evaluation metrics.

Jupyter Notebook / VS Code: Development environments for writing, testing, and debugging code.

Optional / Future Tools: Power BI, Plotly, or Dash for advanced interactive visualizations and dashboards.

CHAPTER 2: LITERATURE REVIEW/BACKGROUND STUDY

2.1 Existing Systems

There have been numerous systems and platforms evolved over time to aid machine learning, data analysis and visualization. Though these tools are helpful, each possesses its own strengths and weaknesses:

WEKA (Waikato Environment for Knowledge Analysis):

One of the most popular open source data mining tools that has a graphical user interface allowing access to a number of ML algorithms, data preprocessing and visualization possibilities.

Supports classification, regression, clustering and association rule discovery.

Limitations: Less interactive visualization, mostly manual algorithm selection, and limited iterative experimentation.

RapidMiner:

ML provides drag-and-drop workflow construction for data preprocessing, model training, and evaluation.

Supports sophisticated ML methods and offers rich analysis environment.

Limitations: Resource hungry, slow with big data sets, and can be complex for new users.

orange:

Open source platform emphasizing ease of use with visual programming for ML and data analysis.

Can be used to build workflows using widgets, analyze data interactively and visualize outcomes.

Drawbacks: Provides less flexibility to optimize models using complex model optimization and parameter tuning.

Python-based ML Notebook (Jupyter, Google Colab):

Total Python library access including Pandas, NumPy, Matplotlib, Seaborn and Scikit-Learn.

Grants complete flexibility to build, train, and test models.

Drawbacks: More programming-oriented, which is less inviting for new users or non-technical users.

2.2 Limitations of Existing Systems

These tools are not without a few disadvantages, however, even with their robust machine learning features:

steep learning curve: Beginners will struggle to understand concepts of ML, deal with library dependencies, and write code.

Limited interactivity: It's difficult to navigate and fully understand the data because a lot of the visualizations are static.

Manual workflow: Preprocessing, model training, evaluation, and comparison are some examples of multi-step and time-consuming activities.

Limited accessibility: Tools are not suitable for real-time analysis, may need licensing, or consume resources.

Fragmentation of workflow: In order to complete ML activities, users often need to move in between multiple tools, which reduces efficiency.

These limitations point towards the need for an integrated, user-friendly platform that supports interactive, end-to-end machine learning processes.

2.3 Proposed System Overview

The proposed system is a GUI-based tool for dataset analysis, visualization and machine learning, designed to address the limitations of existing platforms. Key components and features include:

User friendly GUI:

The system offers a simple and intuitive interface that reduces the need for coding. Users can perform data analysis and machine learning tasks with minimal technical expertise.

Dataset support and preprocessing:

Users can upload datasets in CSV or Excel format. The tool automatically handles missing values, codes categorical data, measures numerical functions and provides options for feature selection.

The platform creates interactive visualizations for exploratory data analysis (EDA), including pie charts, scatter plots, box plots, histograms, and correlation matrices. Exploring relationships, identifying outliers, and comprehending data distribution are all made simple for users.

Machine learning model integration:

The system supports many supervised ML algorithms for classification and regression, including decision trees, random forests, logistic regression, k-nearest neighbors, support vector machines, and linear/polynomial regression.

Model assessment and comparison:

After training, the model is evaluated using metrics such as precision, accuracy, recall, F1 score, R^2 score, and error metrics. Users can compare performance visually using charts and tables.

Iterative experiment:

The tool allows users to change parameters, retrain models and instantly see changes in model performance, supporting experimentation and learning.

Advantages of the proposed system:

Integrates the entire ML workflow into a single platform.

ML reduces the complexity and effort required for tasks.

Provides interactive visualizations for better understanding of data.

CHAPTER 3: SYSTEM ANALYSIS & DESIGN

3.1 Introduction

System analysis and design includes reading the requirements, knowledge existing workflows, and creating a blueprint to increase an green machine. In this mission, the purpose is to create a GUI-based totally device for dataset evaluation, machine studying model schooling, evaluation, and prediction. The machine is designed to simplify complex ML techniques, making them handy to both technical and non-technical users.

The primary goal is to integrate all important gadget studying duties—statistics preprocessing, exploratory information evaluation, version building, assessment, and prediction—right into a single, interactive, and user-friendly platform. This method lets in customers to recognition on insights as opposed to code and fosters faster experimentation, higher understanding of datasets, and records-pushed decision-making.

3.2 System Architecture

The proposed device is based totally on a modular architecture that separates the ML workflow into distinct, manageable components. Each module plays particular obligations while interacting seamlessly with others to provide an give up-to-give up answer.

Modules of the System:

Data upload and preprocessing module

Users can upload datasets in CSV or Excel layout.

Pretreatment consists of:

Handling of missing values (deletion or imputation)

Categorical variable encoding (One-Hot, Label Encoding)

Feature scaling (standardization or normalization)

Feature choice based on correlation or significance measurements

This guarantees that the dataset is easy, constant and ready for evaluation.

Exploratory Data Analysis (EDA) module

Provides interactive visualizations which include histogram, scatter plot, box plot and correlation matrix.

Statistical summaries (suggest, median, mode, preferred deviation, variance) are displayed.

Allows customers to identify styles, discover outliers and apprehend relationships among variables earlier than training fashions.

Model choice and education module

Provides a lot of supervised ML algorithms:

Classification: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine

Regression: Linear regression, polynomial regression, random wooded area regression

Enables parameter setting for every version (hyperparameters consisting of intensity, gaining knowledge of price, neighbors, and many others.).

Efficiently trains decided on models and shops them for assessment and prediction.

Model evaluation and contrast module

Evaluates trained models using numerous performance metrics:

Accuracy, precision, consider, F1 score for class

R^2 rating, imply squared error, root mean squared mistakes of regression

Visual representation of overall performance through bar charts, line graphs, confusion matrices, and different plots.

Supports comparing a couple of models on the equal time to choose the version with the nice performance.

3.3 System Flow / Data Flow

Step-via-Step Workflow:

User uploads dataset →

Dataset passes thru preprocessing →

Cleaned statistics is explored thru EDA →

ML models are selected and trained →

Model evaluation and evaluation →

Prediction on new/unseen facts →

Results visualized and supplied to consumer.

Data Flow Diagram (DFD Level zero – Context Level):

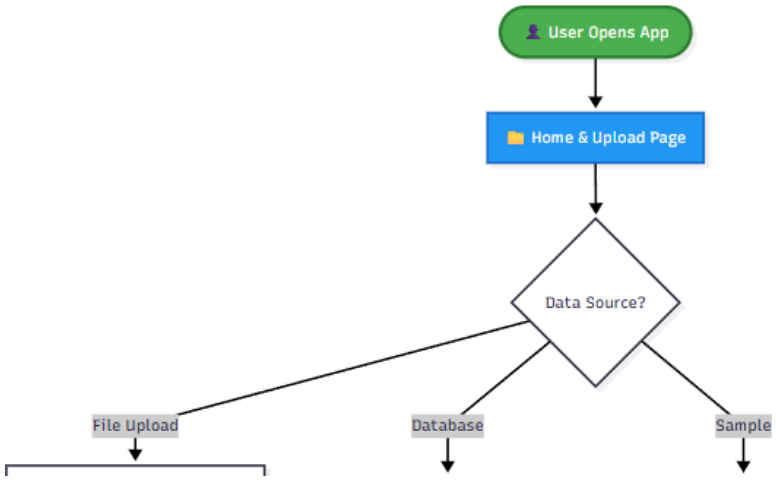
External Entity: User

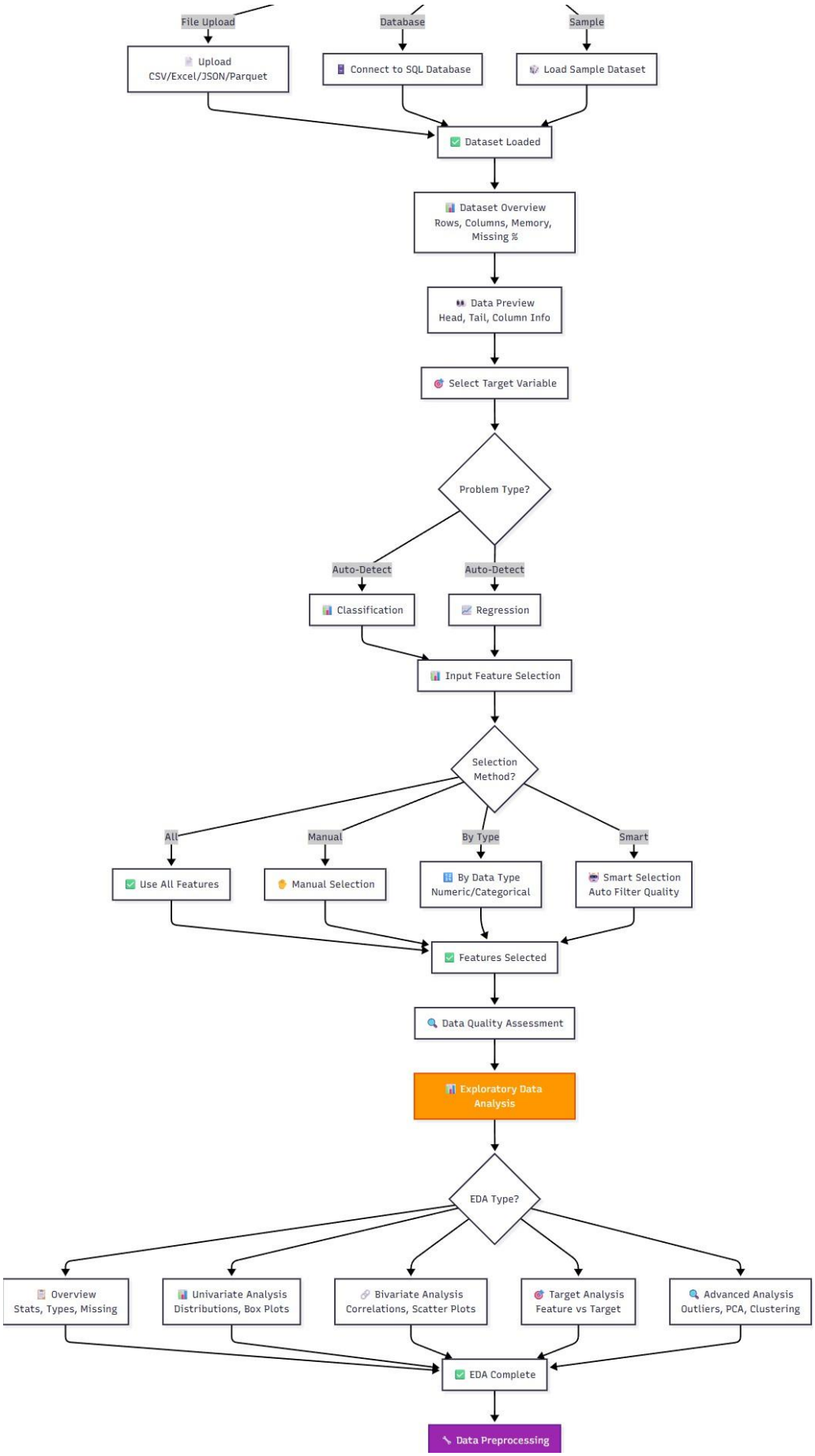
Processes: Data Upload, Preprocessing, EDA, Model Training, Evaluation, Prediction

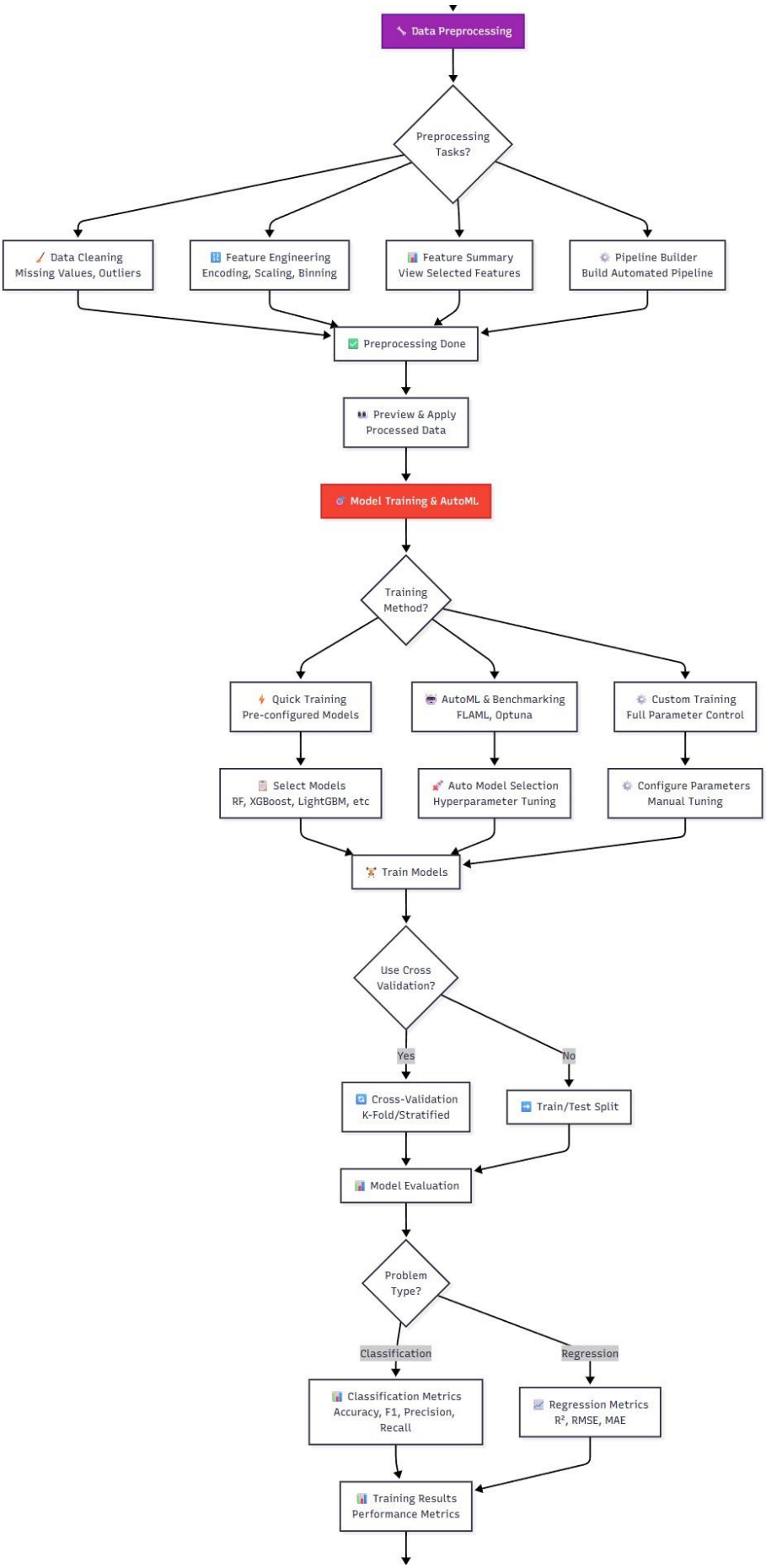
Data Stores: Dataset Storage, Model Storage

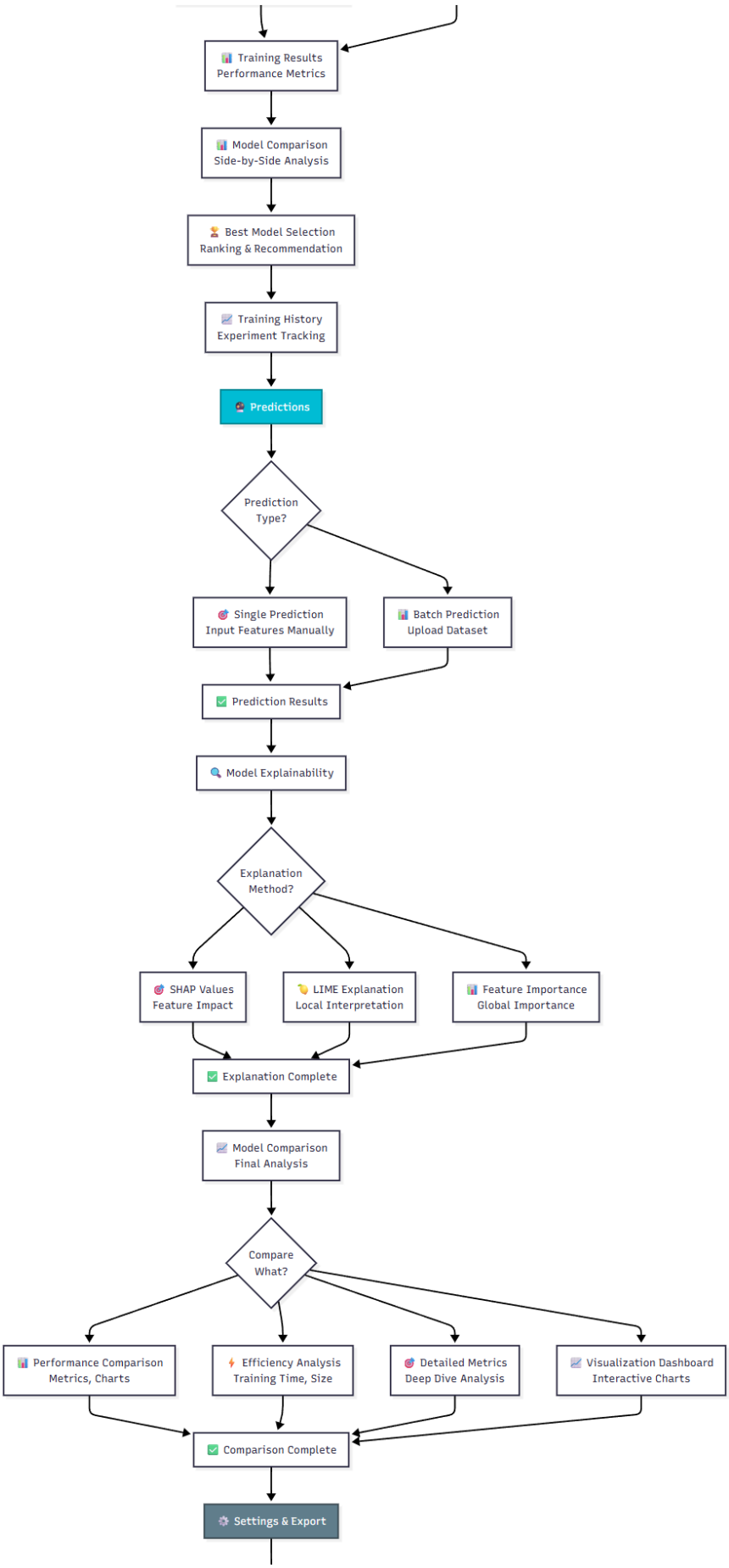
Flow: User inputs → System techniques → Outputs consequences

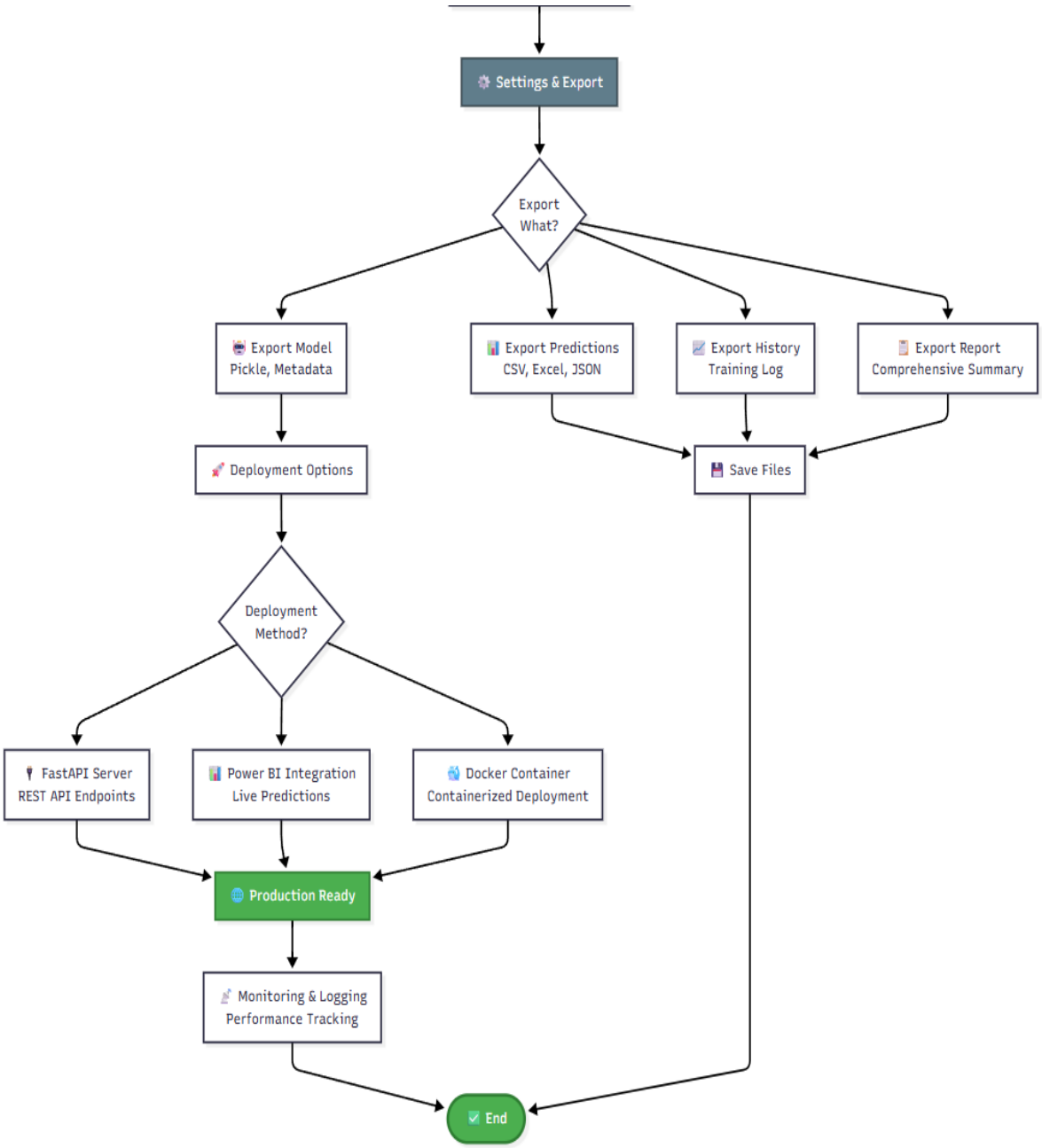
❖ **WORKFLOW:**




File Upload
Database
Sample
Upload CSV/Excel/JSON/Parquet
Connect to SQL Database
Load Sample Dataset
Dataset Loaded
Dataset Overview Rows, Columns, Memory, Missing %
Data Preview Head, Tail, Column Info
Select Target Variable
Problem Type?
Auto-Detect
Auto-Detect
Classification
Regression
Input Feature Selection
Selection Method?
All
Manual
By Type
Smart
Use All Features
Manual Selection
By Data Type Numeric/Categorical
Smart Selection Auto Filter Quality
Features Selected
Data Quality Assessment
Exploratory Data Analysis
EDA Type?
Overview Stats, Types, Missing
Univariate Analysis Distributions, Box Plots
Bivariate Analysis Correlations, Scatter Plots
Target Analysis Feature vs Target
Advanced Analysis Outliers, PCA, Clustering
EDA Complete
Data Preprocessing


Data Preprocessing
Preprocessing Tasks?
Data Cleaning
Missing Values, Outliers
Feature Engineering
Encoding, Scaling, Binning
Feature Summary
View Selected Features
Pipeline Builder
Build Automated Pipeline
Preprocessing Done
Preview & Apply
Processed Data
Model Training & AutoML
Training Method?
Quick Training
Pre-configured Models
AutoML & Benchmarking
FLAML, Optuna
Custom Training
Full Parameter Control
Select Models
RF, XGBoost, LightGBM, etc
Auto Model Selection
Hyperparameter Tuning
Configure Parameters
Manual Tuning
Train Models
Use Cross Validation?
Yes
No
Cross-Validation
K-Fold/Stratified
Train/Test Split
Model Evaluation
Problem Type?
Classification
Regression
Classification Metrics
Accuracy, F1, Precision, Recall
Regression Metrics
R², RMSE, MAE
Training Results
Performance Metrics


📊 Training Results
Performance Metrics
📊 Model Comparison
Side-by-Side Analysis
🏆 Best Model Selection
Ranking & Recommendation
📈 Training History
Experiment Tracking
🔮 Predictions
Prediction Type?
🎯 Single Prediction
Input Features Manually
📊 Batch Prediction
Upload Dataset
✅ Prediction Results
🔍 Model Explainability
Explanation Method?
🎯 SHAP Values
Feature Impact
🍋 LIME Explanation
Local Interpretation
📊 Feature Importance
Global Importance
✅ Explanation Complete
📈 Model Comparison
Final Analysis
Compare What?
📊 Performance Comparison
Metrics, Charts
⚡ Efficiency Analysis
Training Time, Size
🎯 Detailed Metrics
Deep Dive Analysis
📈 Visualization Dashboard
Interactive Charts
✅ Comparison Complete
⚙️ Settings & Export


Settings & Export
Export What?
Export Model Pickle, Metadata
Export Predictions CSV, Excel, JSON
Export History Training Log
Export Report Comprehensive Summary
Deployment Options
Save Files
Deployment Method?
FastAPI Server REST API Endpoints
Power BI Integration Live Predictions
Docker Container Containerized Deployment
Production Ready
Monitoring & Logging Performance Tracking
End

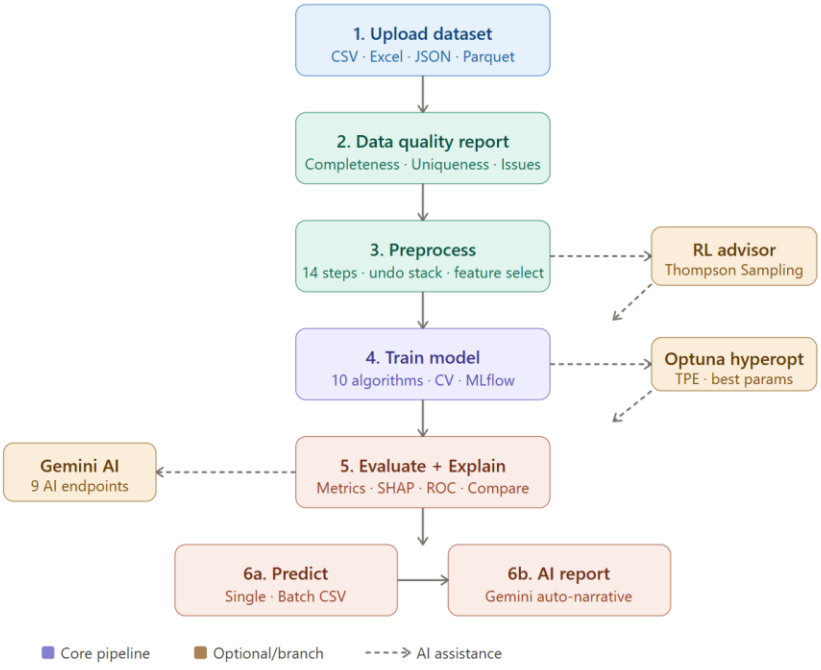


Figure 2 — End-to-end ML pipeline with optional branches

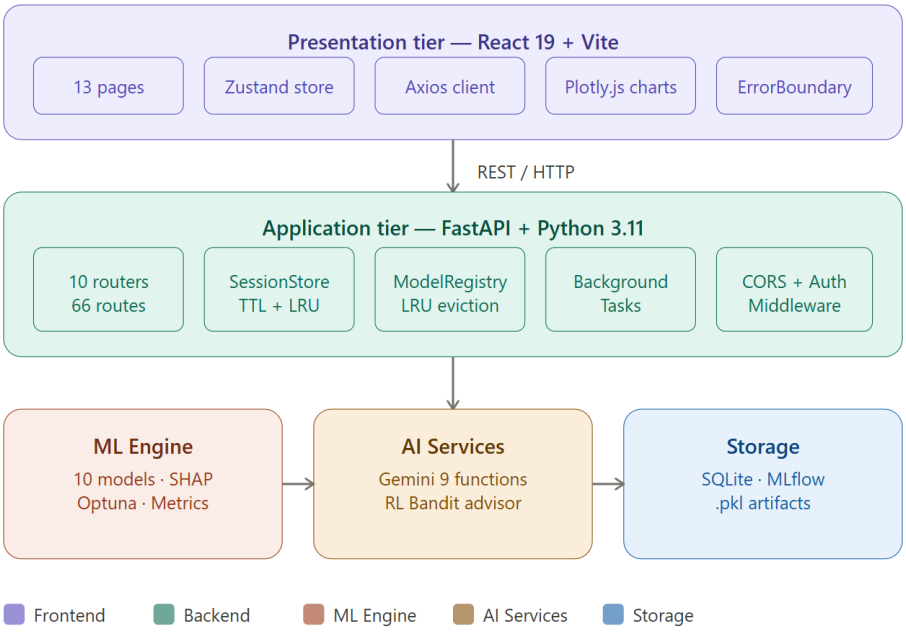


Figure 1 — Three-tier system architecture of the ML Platform

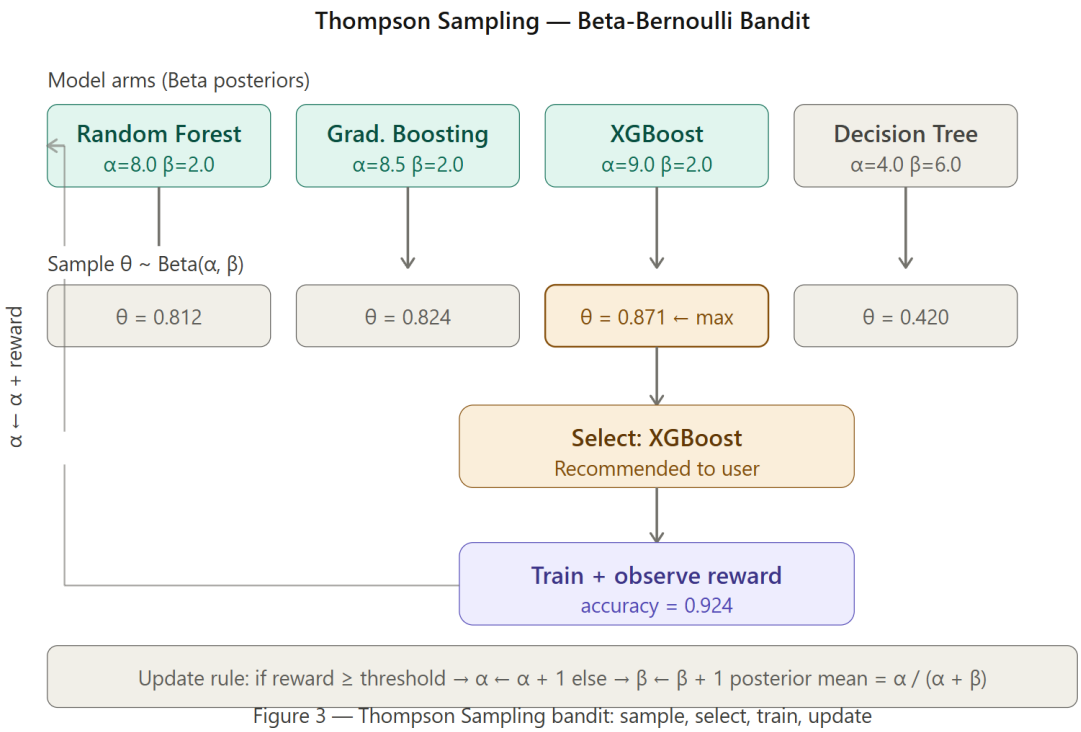


Figure 3 — Thompson Sampling bandit: sample, select, train, update

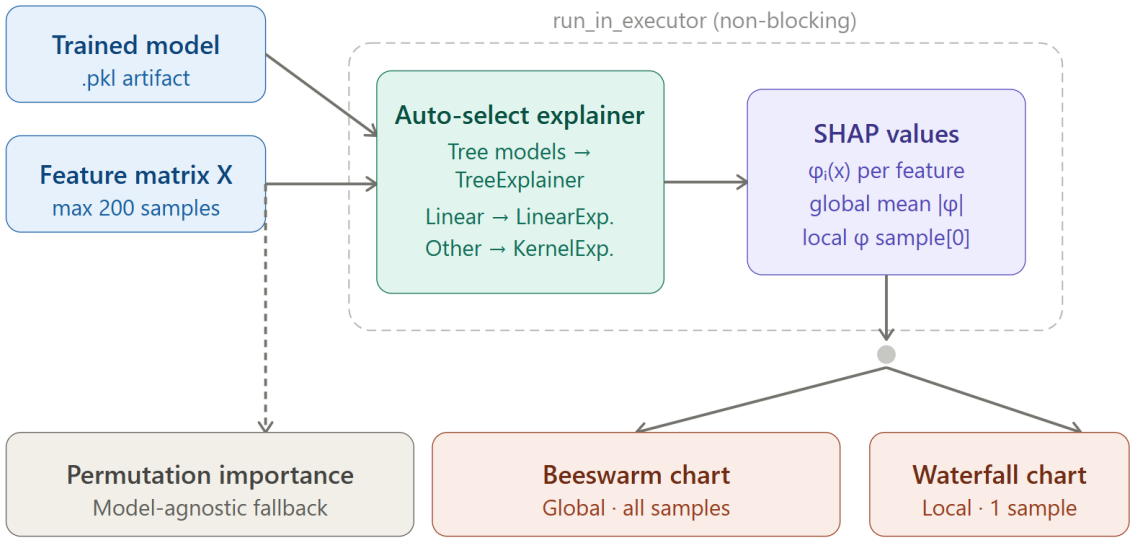


Figure 4 — XAI pipeline: explainer auto-selection, async SHAP computation, dual chart outputs

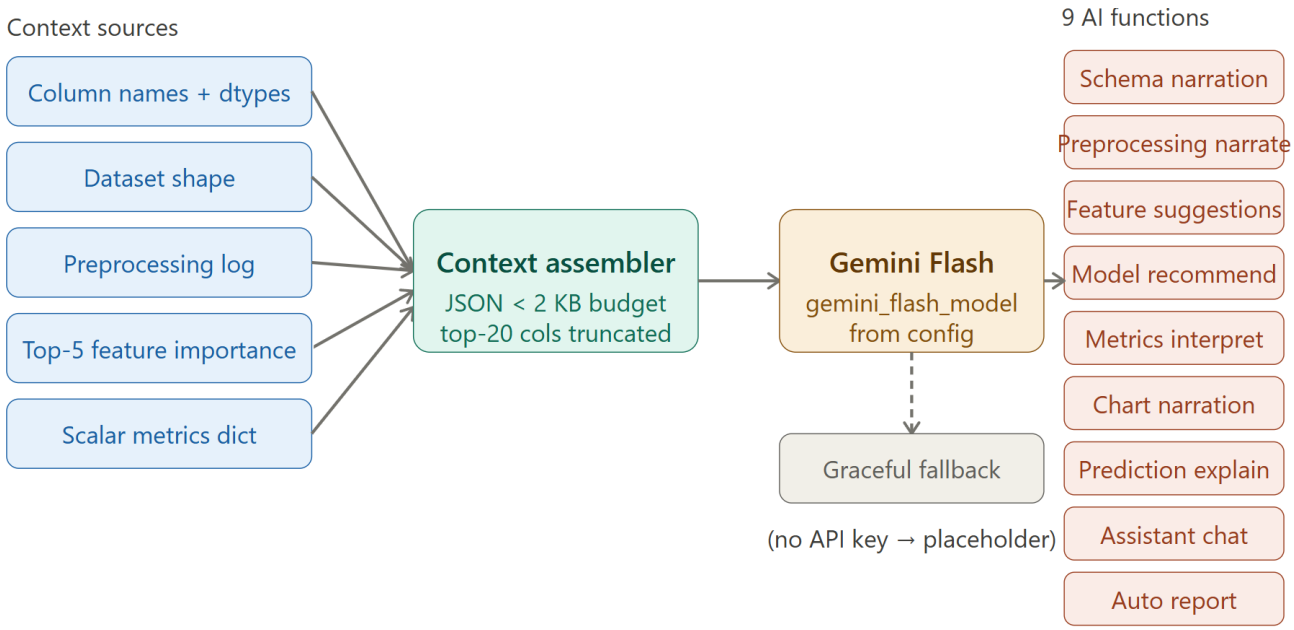


Figure 5 — Gemini AI context pipeline: budget-constrained assembly, 9 functions, graceful degradation

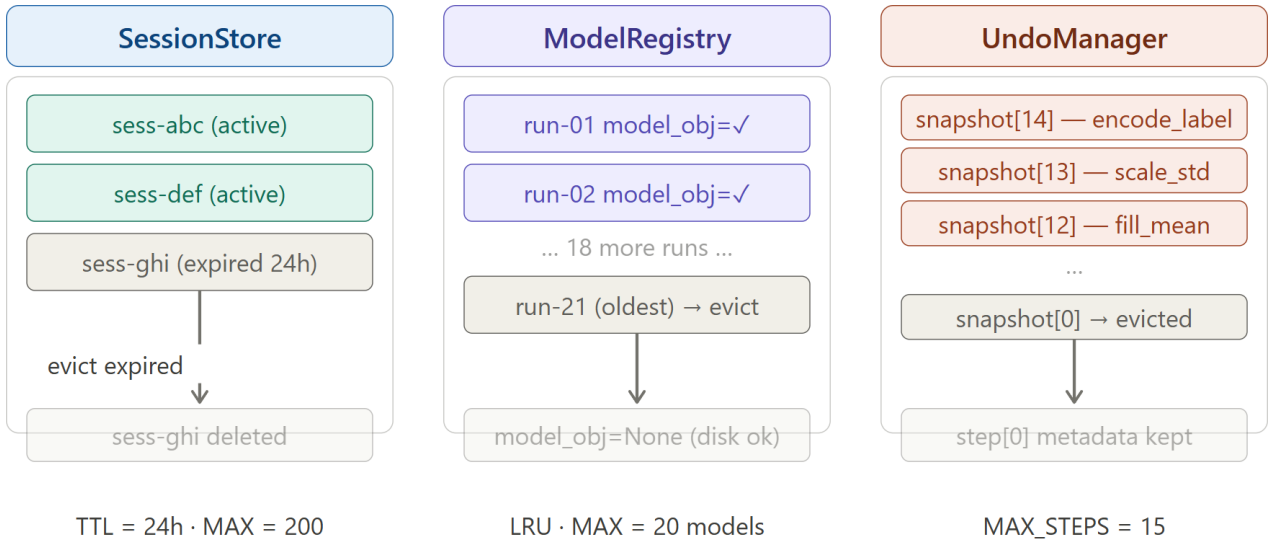


Figure 6 — Three memory management mechanisms preventing unbounded RAM growth

3.4 Module description

Data upload and preprocessing handles dataset upload, cleaning, encoding, scaling and feature selection.

Exploratory data analysis provides interactive visualizations and statistical summaries for data insights.

Model selection and training provides multiple ML algorithms, trains models and allows parameter tuning.

Model Evaluation and Comparison evaluates models using calculations and graphs; Helps to choose the best model.

The forecasting module generates forecasts on new or unseen data using the trained model**.**

3.5 Design Considerations

Ease of use: The GUI is designed to be simple, intuitive and visually appealing, lowering the studying length for novices.

Scalability: The modular layout makes it possible to add extra algorithms, visualizations and advanced ML strategies in the future.

Interactivity: Users can dynamically interact with datasets and fashions, exchange parameters and examine outcomes in real time.

Accessibility: Minimal coding is needed, making it suitable for students, researchers and experts without technical understanding.

Performance: green handling of medium to massive records sets at the same time as maintaining it of the GUI.

3.6 Tools and technologies utilized in layout

Python: The middle programming language for backend logic.

Streamlight: GUI framework for developing interactive, web-based totally interfaces.

Pandas and NumPy: Data manipulation and numerical calculations.

Matplotlib and Seaborn: Visualization libraries for EDA and performance metrics.

Scikit-examine: Implements ML algorithms, preprocessing strategies, and evaluation metrics.

Jupyter Notebook/VS Code: Development and Test Environments.

3.7 Advantages of the Proposed System Design

End-to-stop workflow in a single platform, decreasing complexity.

Interactive and visually intuitive interface for clean know-how.

Facilitates iterative experimentation, parameter tuning, and model assessment.

Reduces dependency on programming information, making ML handy to all users.

Serves each academic and realistic functions, supporting customers understand and apply ML standards.

3.8 Summary

The proposed system design ensures a continuing drift of information from add to prediction. By combining modularity, interactivity, and person-centric design, the device simplifies gadget learning workflows and makes superior analytics available to a much broader audience. Its based approach allows customers to explore datasets, teach fashions, compare performance, and generate predictions successfully, making it a complete, academic, and practical tool for actual-international packages.

CHAPTER 4: Implementation Details

4.1. Functions

User-pleasant interface: Streamlit gives an intuitive GUI for easy navigation and interplay.

Data add and management: Users can upload datasets in codecs inclusive of CSV, Excel or TXT.

Data visualization: Supports multiple charts-line, bar, scatter, histogram, pie for short insights.

Data Analysis Tools: Performs descriptive facts, correlation evaluation, lacking values control and facts cleaning.

Machine learning models: Allows schooling and assessment of fashions such as linear regression, logistic regression, choice trees, random wooded area and KNN.

Model Comparison: Users can evaluate multiple fashions based on accuracy, precision, keep in mind and F1 scores.

Forecasting device: Enables customers to make predictions on new datasets the use of trained models.

Download and export: Results, charts and model reports can be downloaded for further use. Interactive filtering: Users can clear out information in actual time and customise visualizations.

4.2. Functionality

Dataset Handling: Reads datasets, previews information, checks for null values, and handles missing records mechanically.

exploratory Data Analysis (EDA): Generates statistical summaries and plots to understand traits and distributions.

Model Training: Offers options to pick features, goal variables, split records, and educate ML models.

Evaluation Metrics: Calculates and shows metrics like accuracy, confusion matrix, ROC-AUC, and so forth.

Dynamic Updates: Changes in input or parameters proper away reflect in visualizations and results.

User Guidance: Provides tooltips, mistakes messages, and instructions for easy interplay.

4.3. Module Descriptions

Data Upload Module: Handles importing, studying, and easy validation of datasets.

Preprocessing Module: Cleans statistics, handles missing values, encodes unique capabilities, and scales statistics if wanted.

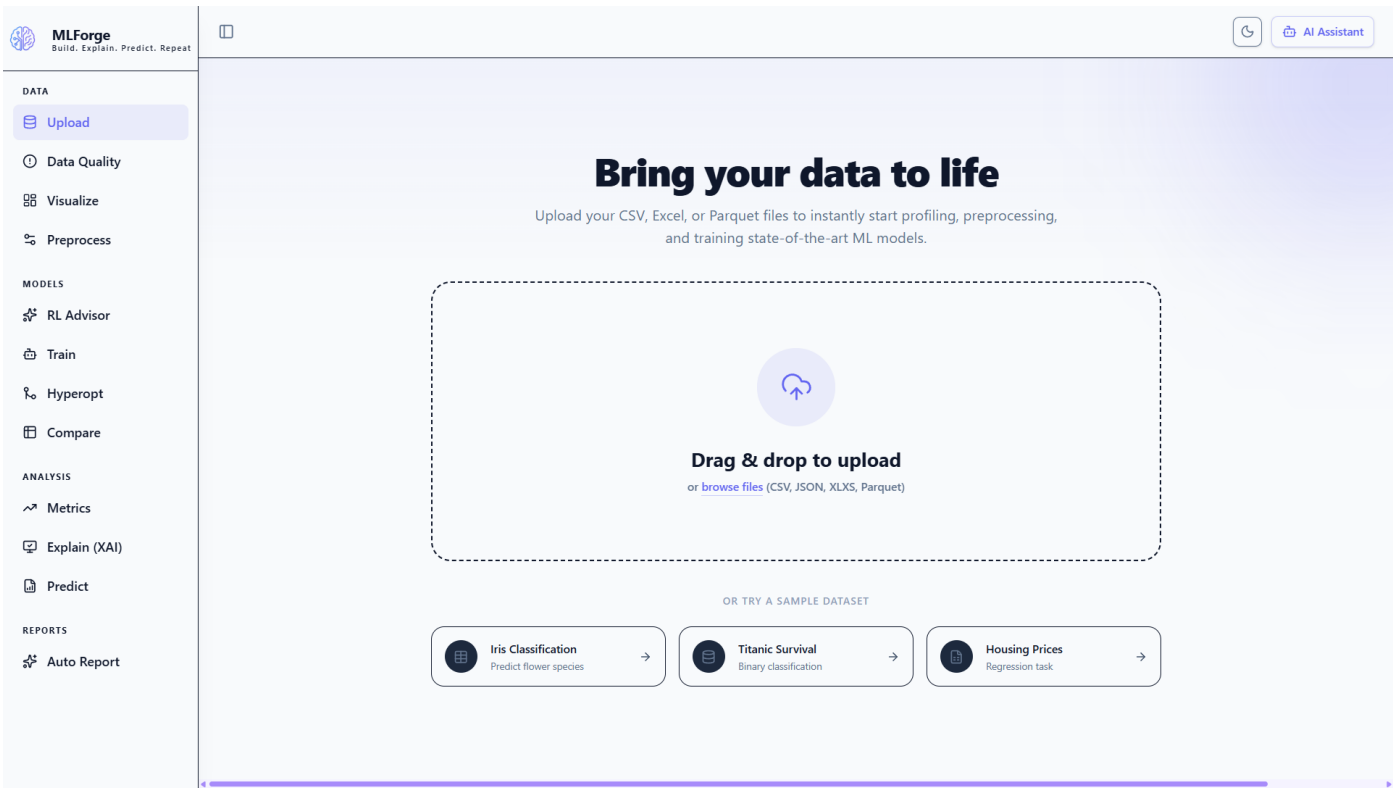
Visualization Module: Generates plots/charts using Matplotlib, Seaborn, and Plotly for interactive exploration.

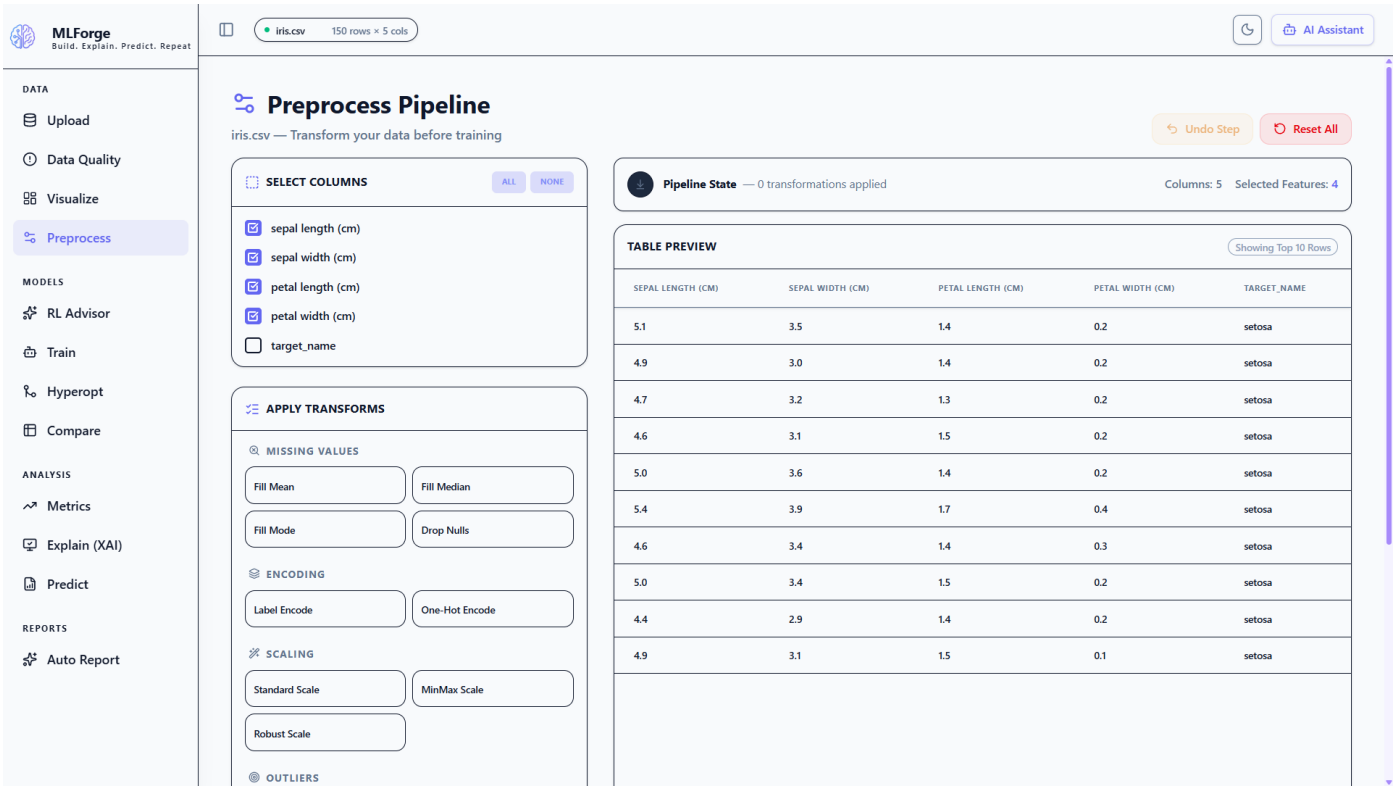
Modeling Module: Implements gadget analyzing algorithms for schooling, finding out, and prediction.

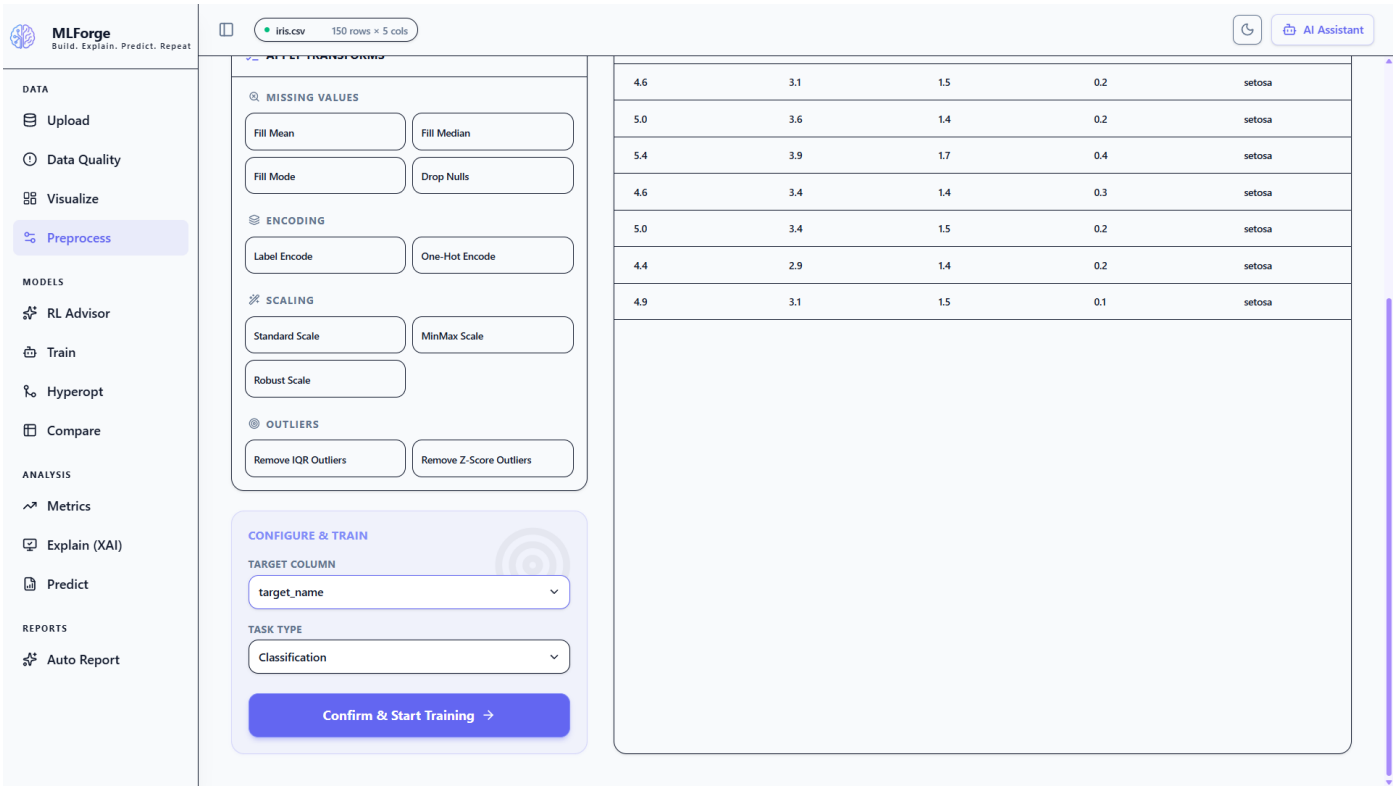
Evaluation Module: Computes version typical performance metrics and visualizes outcomes for evaluation.

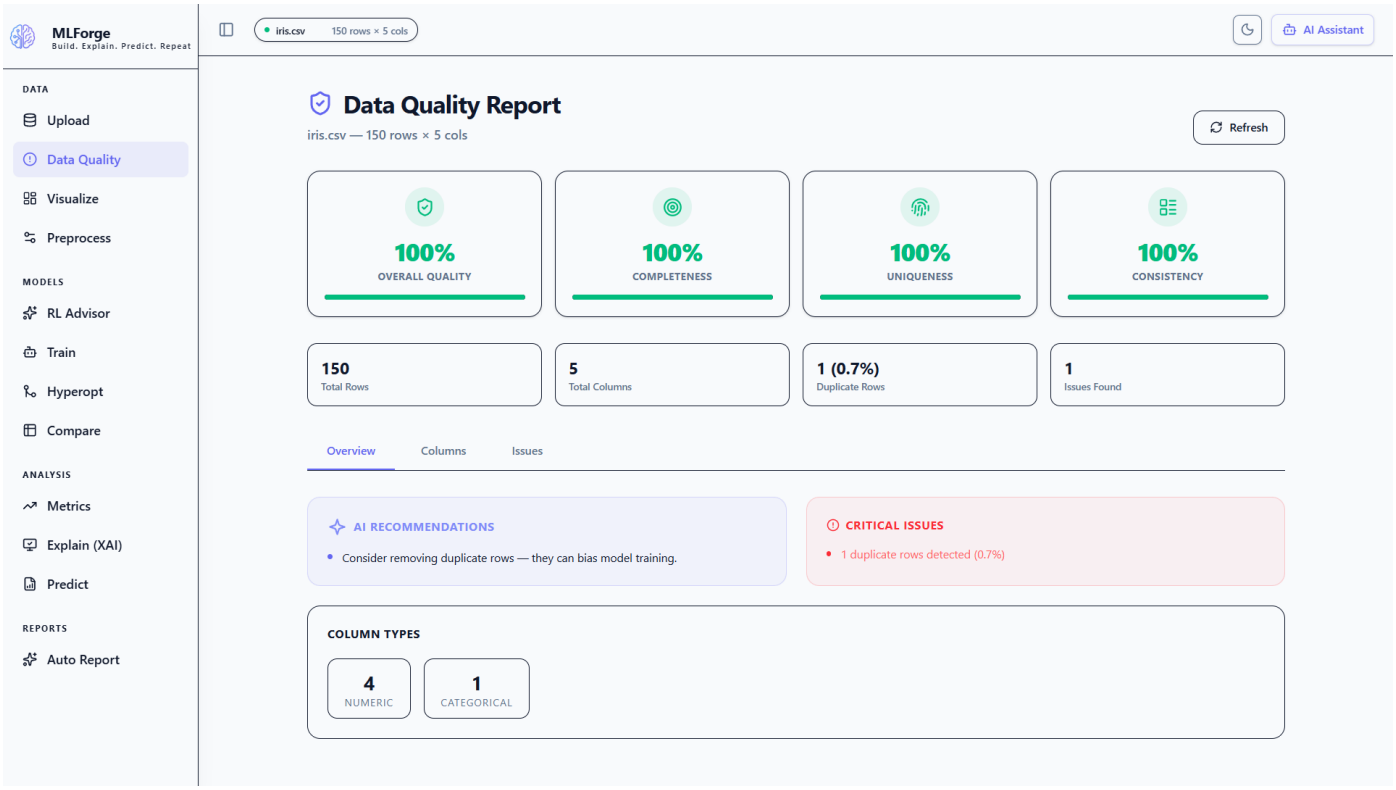
Prediction Module: Uses educated models to expect consequences on new input records.

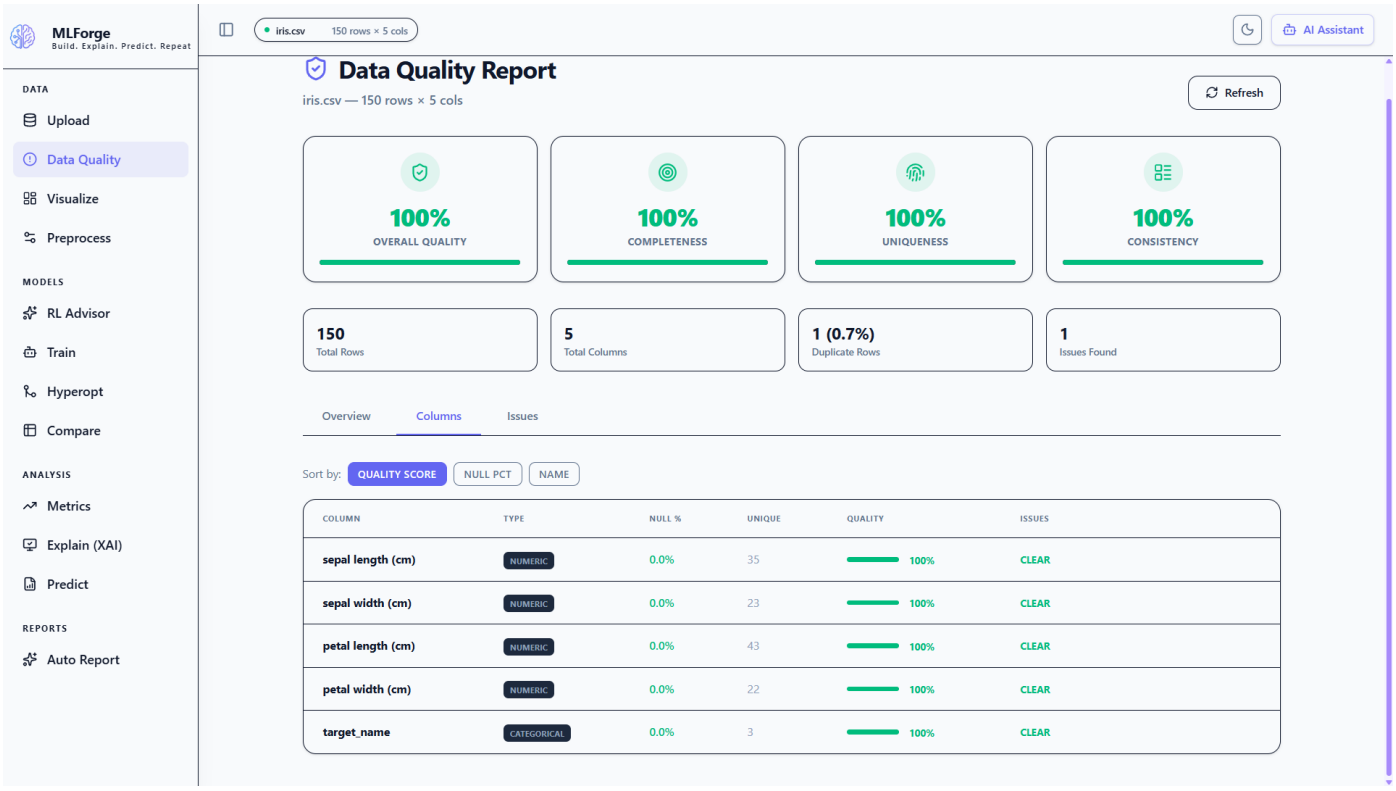
Export Module: Allows customers to download critiques, charts, and prediction outputs in CSV or photo codecs.

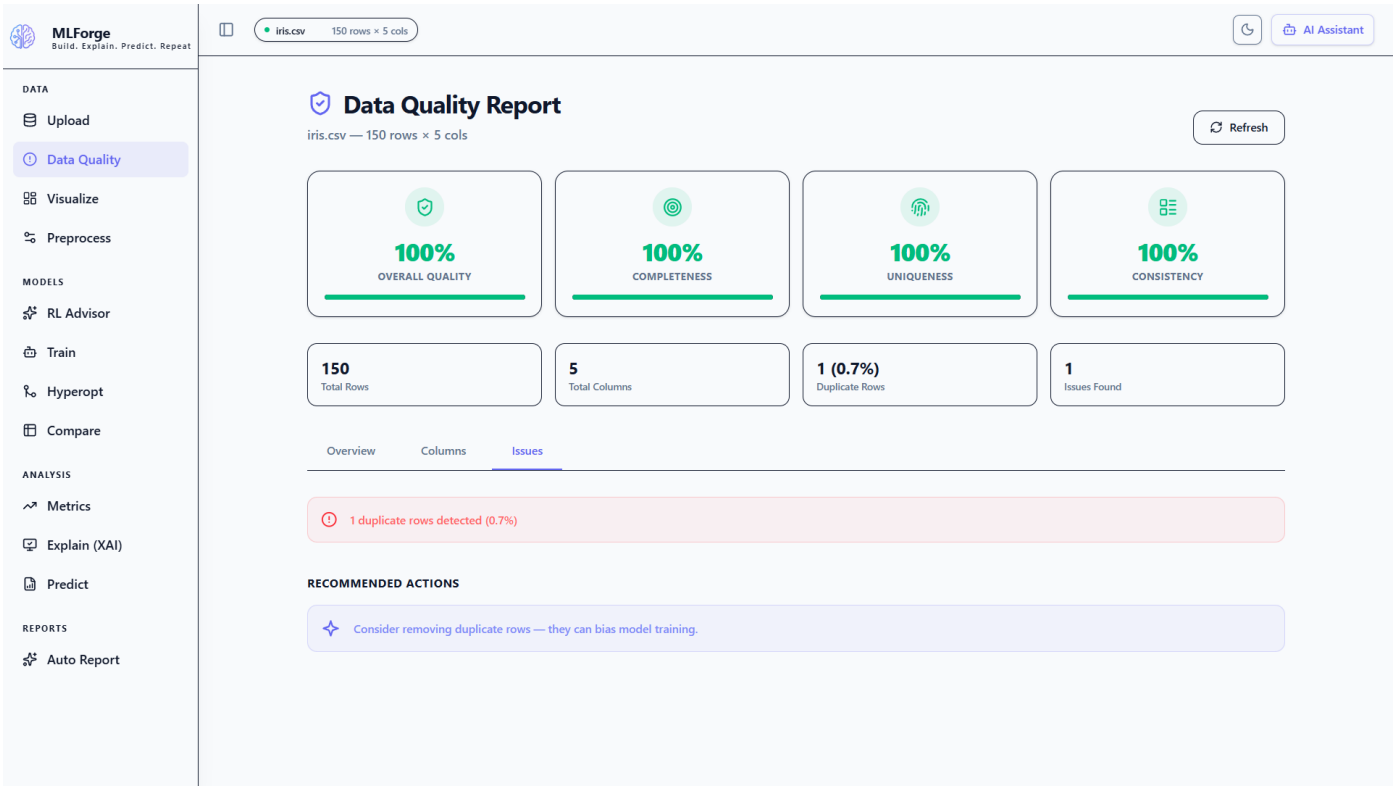

MLForge
Build. Explain. Predict. Repeat
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Bring your data to life
Upload your CSV, Excel, or Parquet files to instantly start profiling, preprocessing, and training state-of-the-art ML models.
Drag & drop to upload
or browse files (CSV, JSON, XLXS, Parquet)
OR TRY A SAMPLE DATASET
Iris Classification
Predict flower species
Titanic Survival
Binary classification
Housing Prices
Regression task

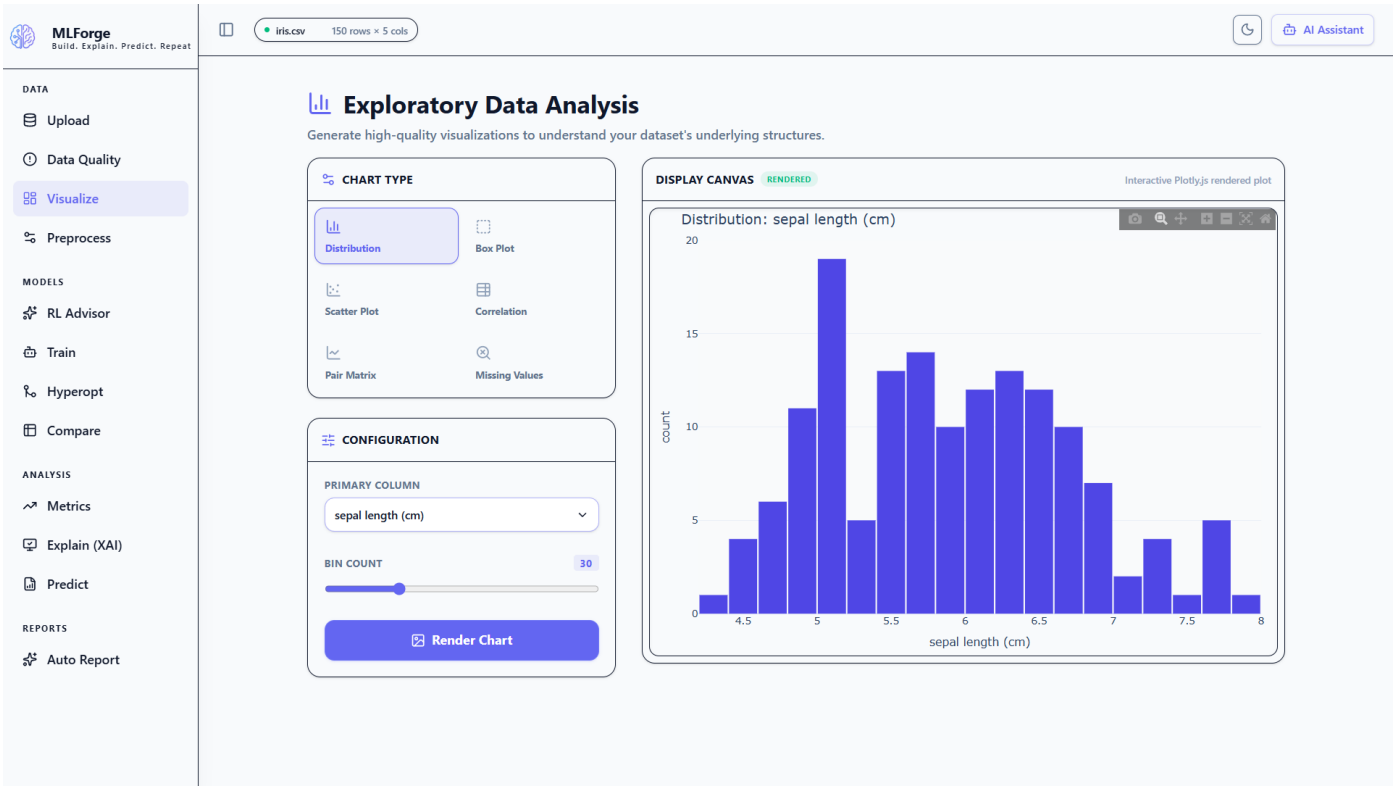

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Preprocess Pipeline
iris.csv — Transform your data before training
Undo Step
Reset All
SELECT COLUMNS
ALL
NONE
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)
target_name
Pipeline State — 0 transformations applied
Columns: 5 Selected Features: 4
APPLY TRANSFORMS
MISSING VALUES
Fill Mean
Fill Median
Fill Mode
Drop Nulls
ENCODING
Label Encode
One-Hot Encode
SCALING
Standard Scale
MinMax Scale
Robust Scale
OUTLIERS
TABLE PREVIEW
Showing Top 10 Rows
SEPAL LENGTH (CM) SEPAL WIDTH (CM) PETAL LENGTH (CM) PETAL WIDTH (CM) TARGET_NAME
5.1 3.5 1.4 0.2 setosa
4.9 3.0 1.4 0.2 setosa
4.7 3.2 1.3 0.2 setosa
4.6 3.1 1.5 0.2 setosa
5.0 3.6 1.4 0.2 setosa
5.4 3.9 1.7 0.4 setosa
4.6 3.4 1.4 0.3 setosa
5.0 3.4 1.5 0.2 setosa
4.4 2.9 1.4 0.2 setosa
4.9 3.1 1.5 0.1 setosa

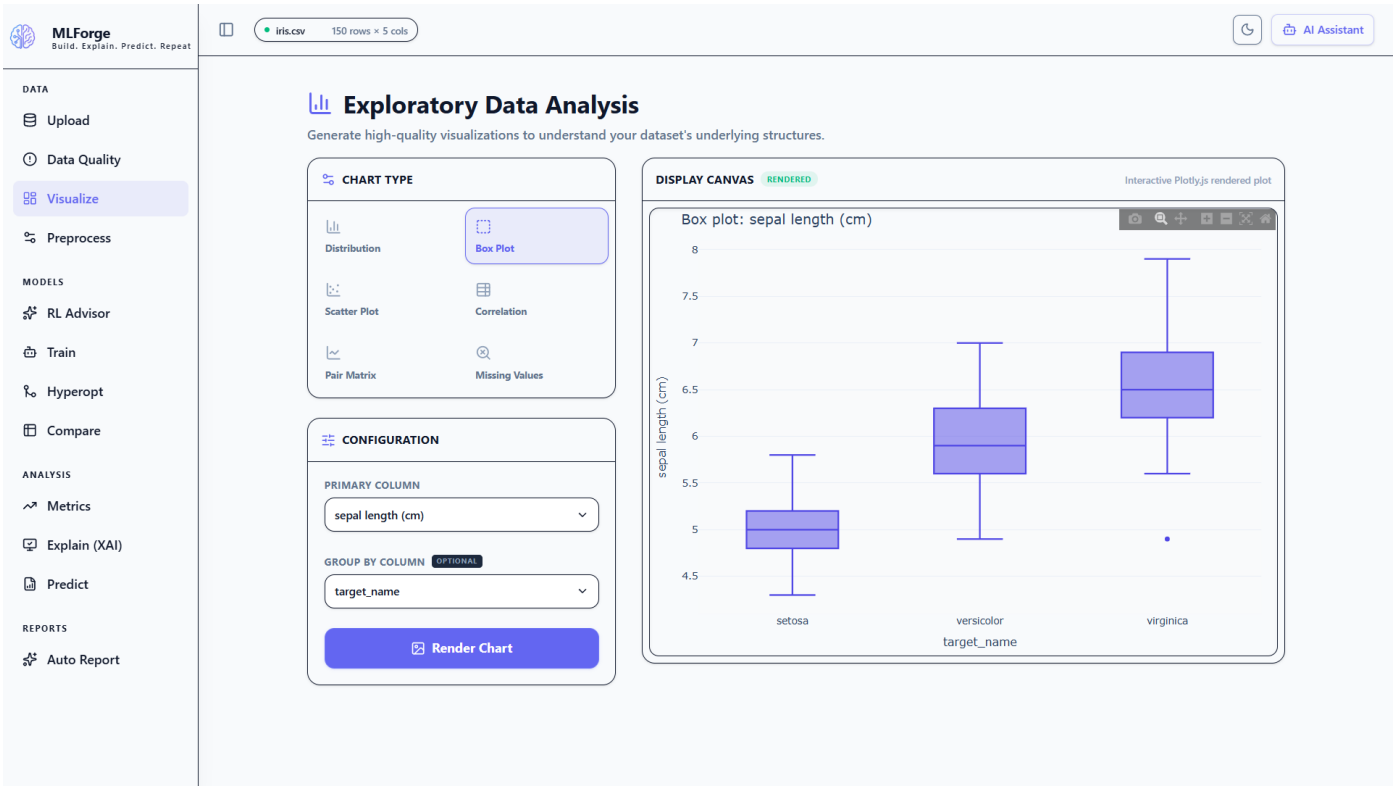

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
MISSING VALUES
Fill Mean
Fill Median
Fill Mode
Drop Nulls
ENCODING
Label Encode
One-Hot Encode
SCALING
Standard Scale
MinMax Scale
Robust Scale
OUTLIERS
Remove IQR Outliers
Remove Z-Score Outliers
CONFIGURE & TRAIN
TARGET COLUMN
target_name
TASK TYPE
Classification
Confirm & Start Training
4.6 3.1 1.5 0.2 setosa
5.0 3.6 1.4 0.2 setosa
5.4 3.9 1.7 0.4 setosa
4.6 3.4 1.4 0.3 setosa
5.0 3.4 1.5 0.2 setosa
4.4 2.9 1.4 0.2 setosa
4.9 3.1 1.5 0.1 setosa

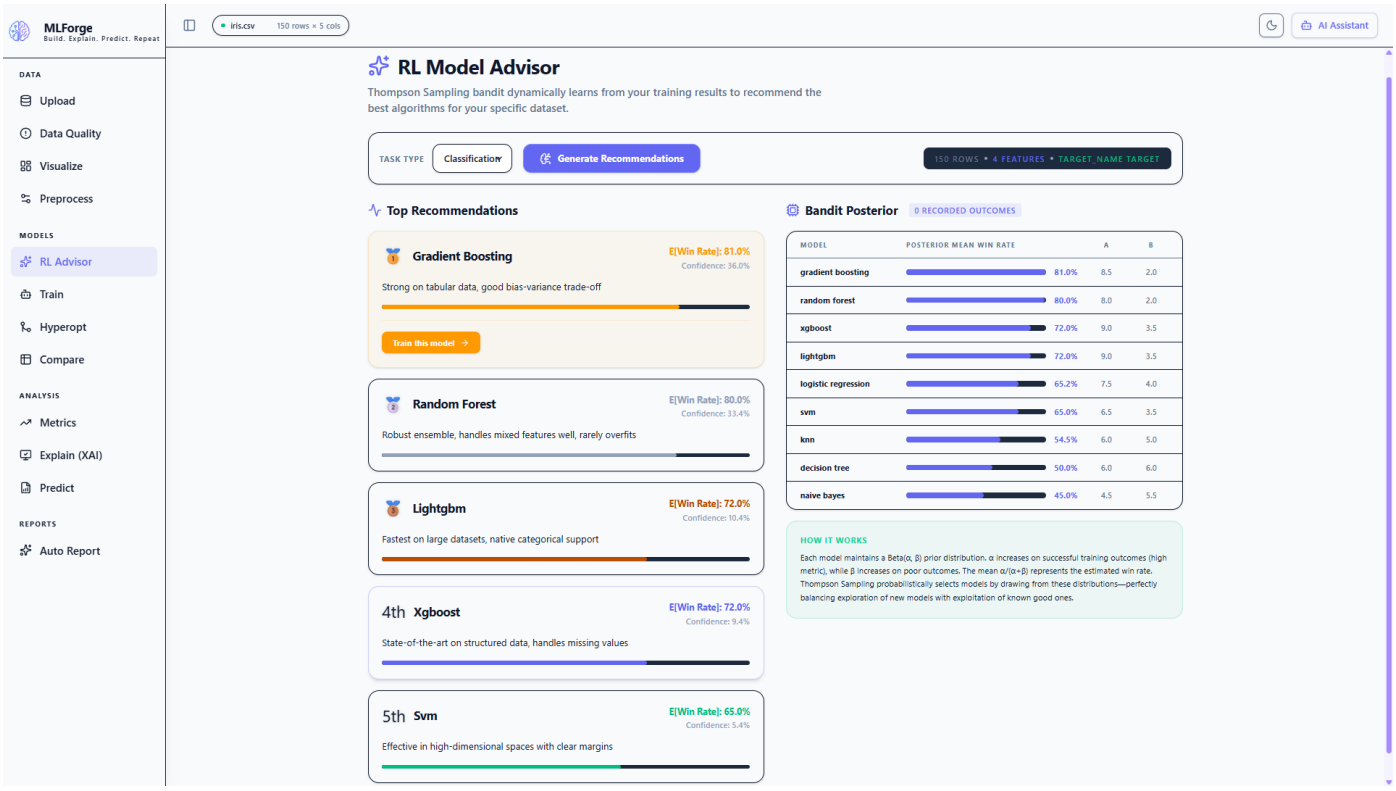

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Data Quality Report
iris.csv — 150 rows × 5 cols
Refresh
100% OVERALL QUALITY
100% COMPLETENESS
100% UNIQUENESS
100% CONSISTENCY
150 Total Rows
5 Total Columns
1 (0.7%) Duplicate Rows
1 Issues Found
Overview
Columns
Issues
AI RECOMMENDATIONS
Consider removing duplicate rows — they can bias model training.
CRITICAL ISSUES
1 duplicate rows detected (0.7%)
COLUMN TYPES
4 NUMERIC
1 CATEGORICAL

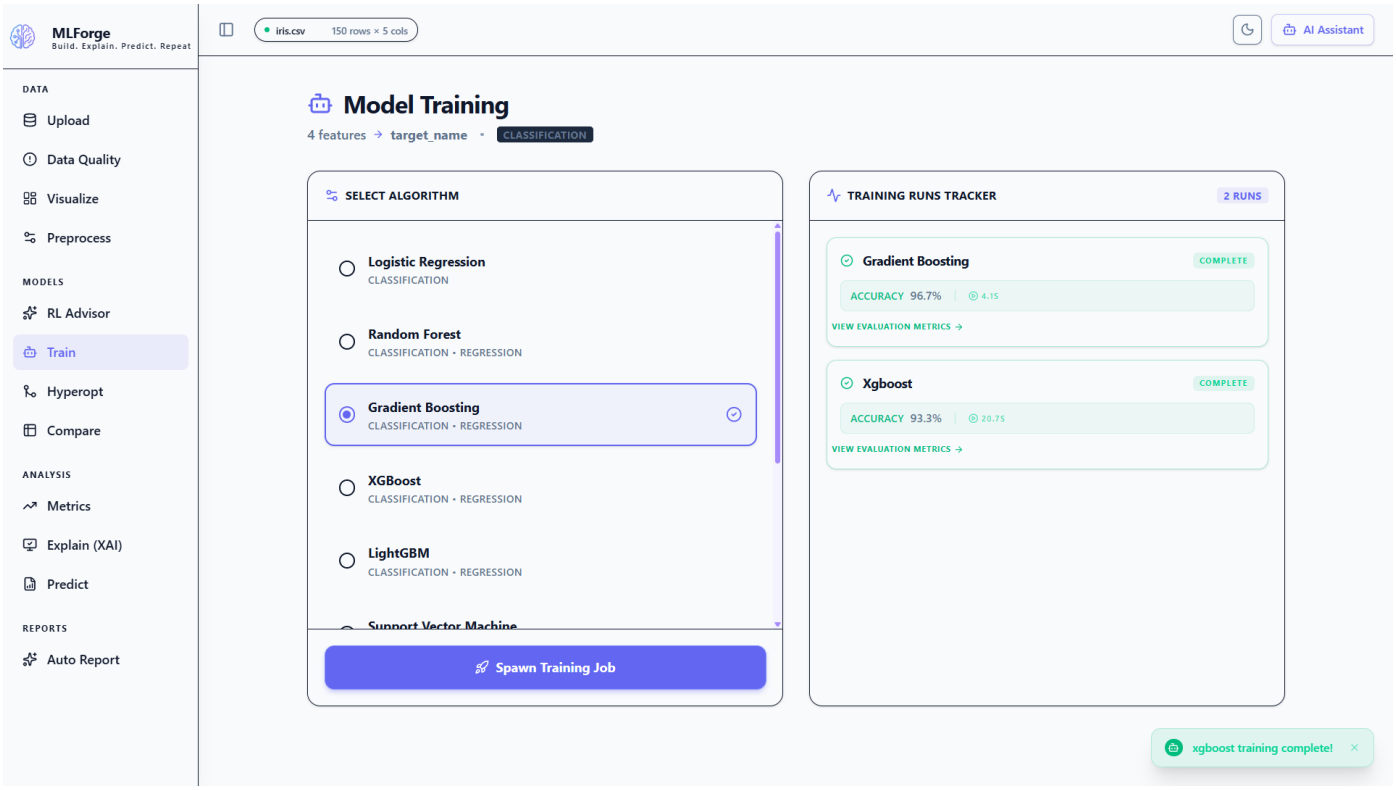

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Data Quality Report
iris.csv — 150 rows × 5 cols
Refresh
100% OVERALL QUALITY
100% COMPLETENESS
100% UNIQUENESS
100% CONSISTENCY
150 Total Rows
5 Total Columns
1 (0.7%) Duplicate Rows
1 Issues Found
Overview
Columns
Issues
Sort by: QUALITY SCORE NULL PCT NAME
COLUMN TYPE NULL % UNIQUE QUALITY ISSUES
sepal length (cm) NUMERIC 0.0% 35 100% CLEAR
sepal width (cm) NUMERIC 0.0% 23 100% CLEAR
petal length (cm) NUMERIC 0.0% 43 100% CLEAR
petal width (cm) NUMERIC 0.0% 22 100% CLEAR
target_name CATEGORICAL 0.0% 3 100% CLEAR

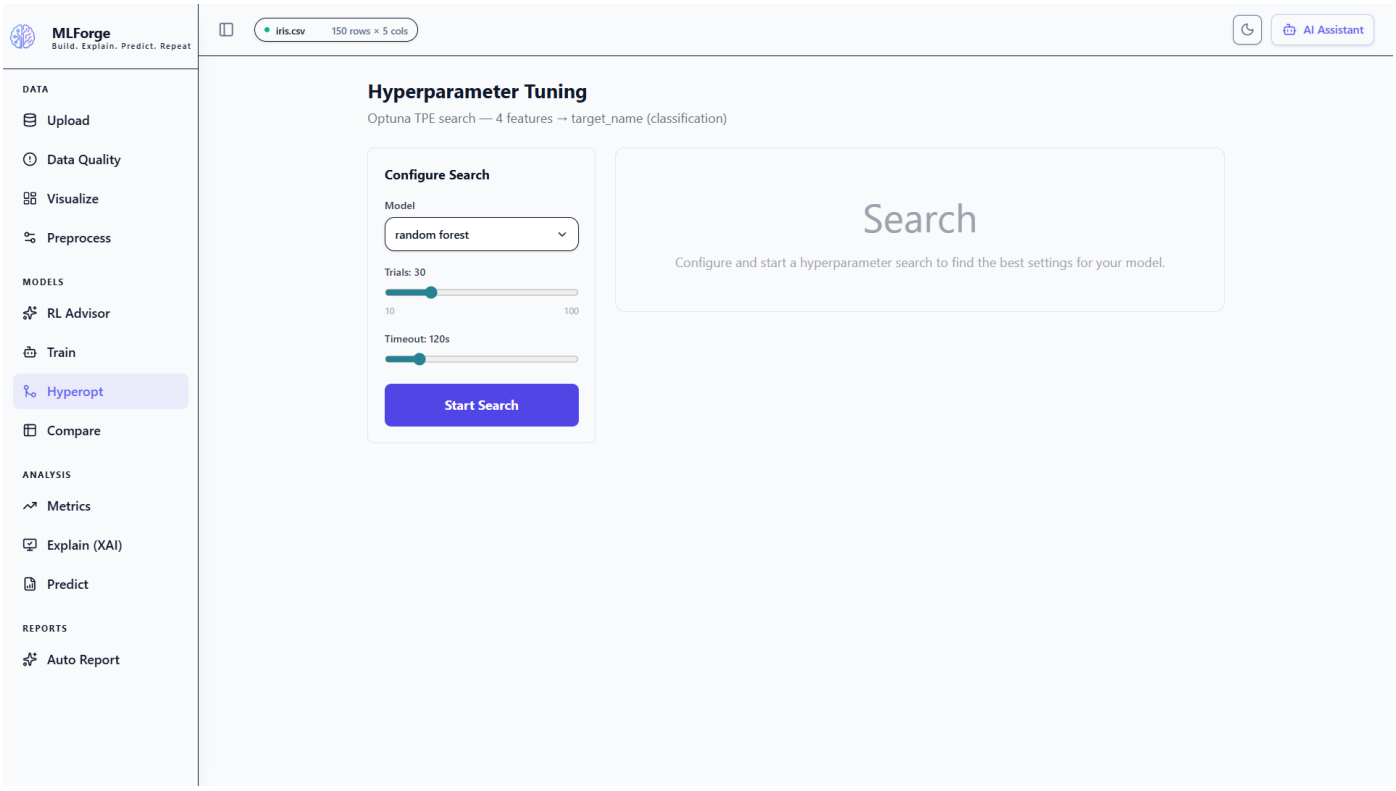

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Data Quality Report
iris.csv — 150 rows × 5 cols
Refresh
100% OVERALL QUALITY
100% COMPLETENESS
100% UNIQUENESS
100% CONSISTENCY
150 Total Rows
5 Total Columns
1 (0.7%) Duplicate Rows
1 Issues Found
Overview
Columns
Issues
1 duplicate rows detected (0.7%)
RECOMMENDED ACTIONS
Consider removing duplicate rows — they can bias model training.

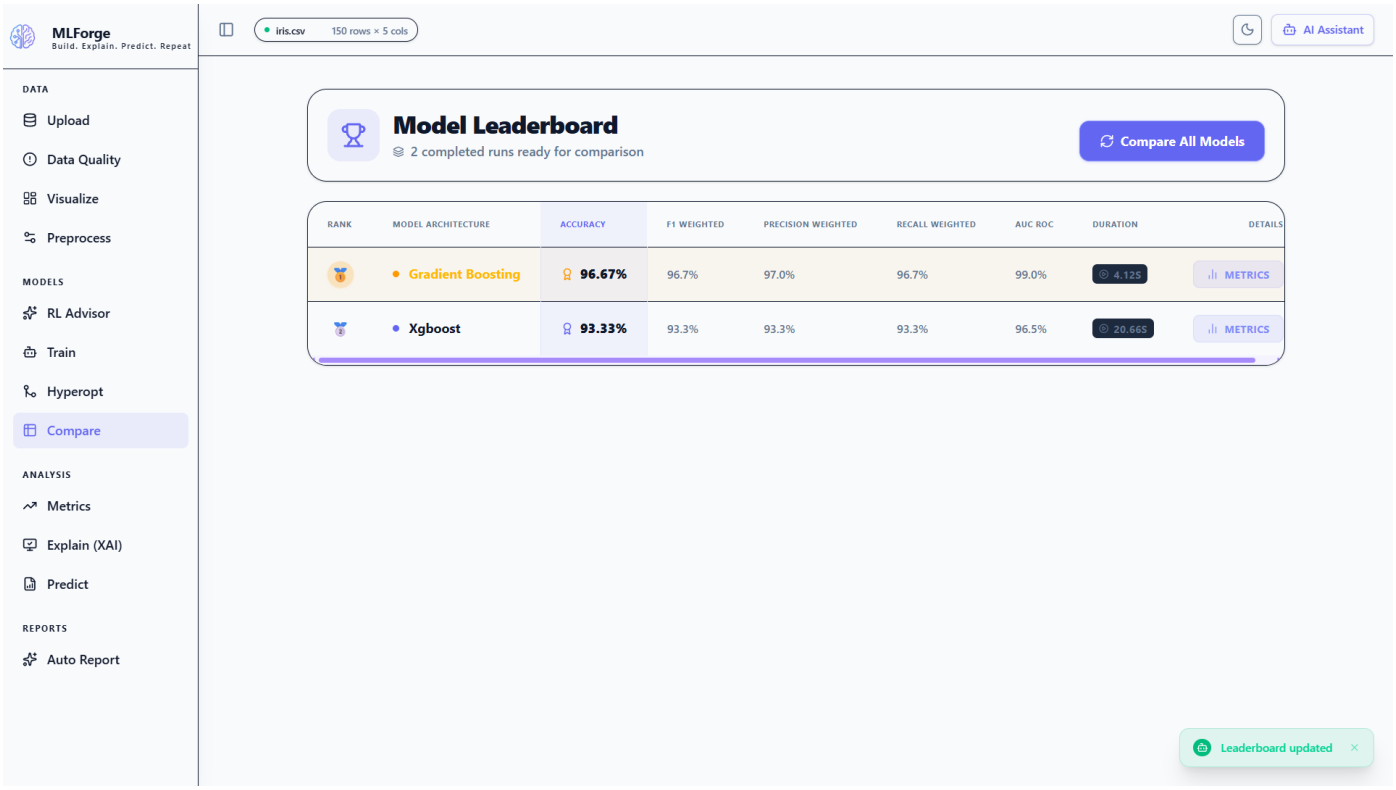

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Exploratory Data Analysis
Generate high-quality visualizations to understand your dataset's underlying structures.
CHART TYPE
Distribution
Box Plot
Scatter Plot
Correlation
Pair Matrix
Missing Values
CONFIGURATION
PRIMARY COLUMN
sepal length (cm)
BIN COUNT
30
Render Chart
DISPLAY CANVAS
RENDERED
Interactive Plotly.js rendered plot
Distribution: sepal length (cm)
count
sepal length (cm)

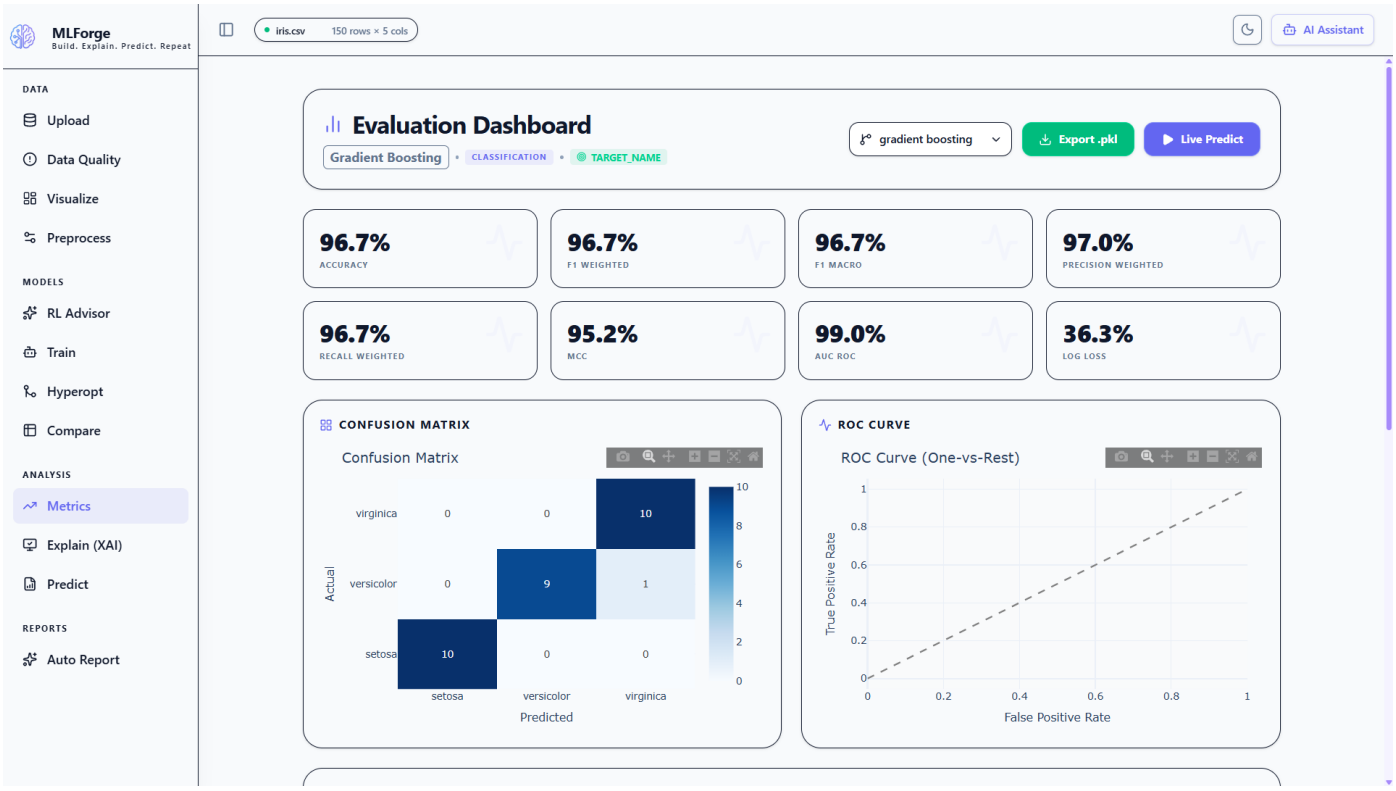

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Exploratory Data Analysis
Generate high-quality visualizations to understand your dataset's underlying structures.
CHART TYPE
Distribution
Box Plot
Scatter Plot
Correlation
Pair Matrix
Missing Values
CONFIGURATION
PRIMARY COLUMN
sepal length (cm)
GROUP BY COLUMN
OPTIONAL
target_name
Render Chart
DISPLAY CANVAS
RENDERED
Interactive Plotly.js rendered plot
Box plot: sepal length (cm)
sepal length (cm)
setosa
versicolor
virginica
target_name

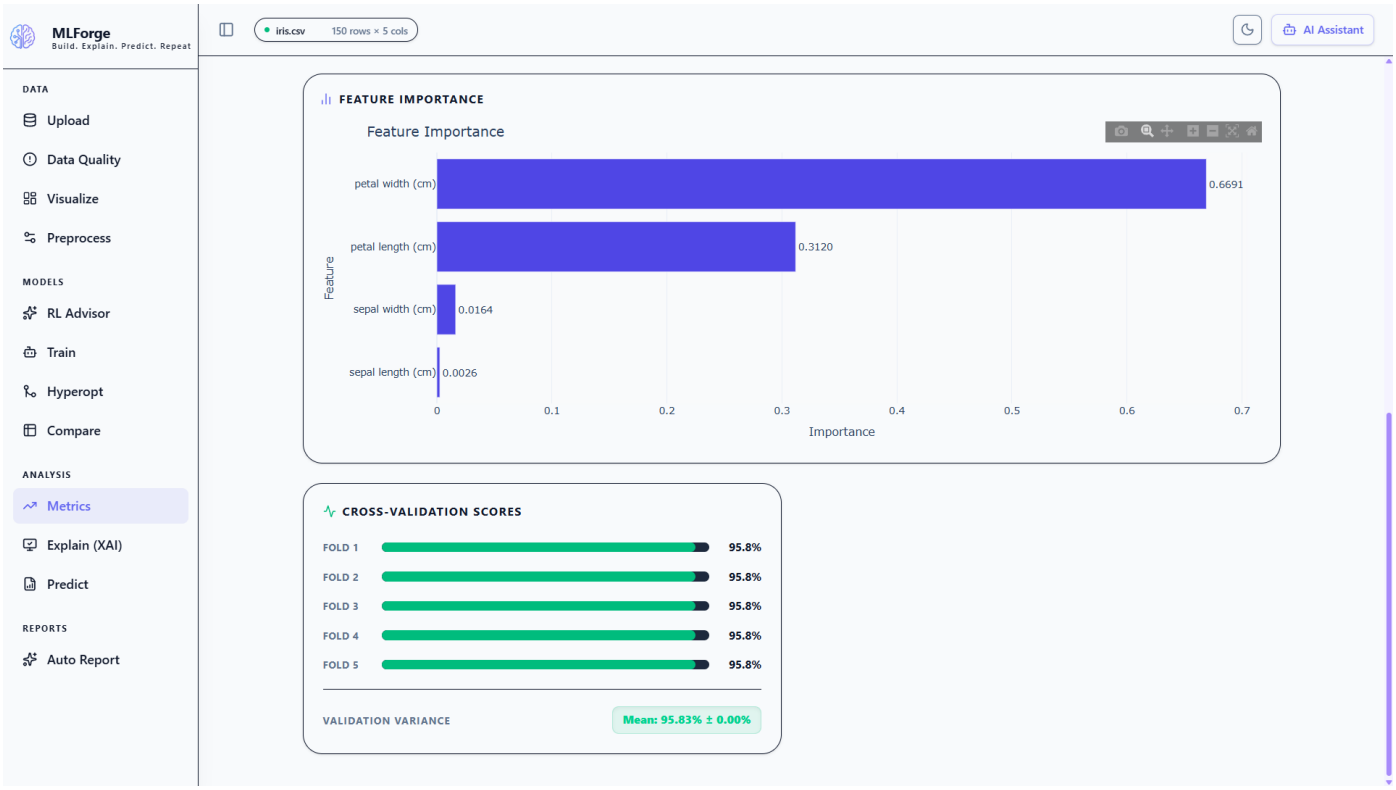

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
RL Model Advisor
Thompson Sampling bandit dynamically learns from your training results to recommend the best algorithms for your specific dataset.
TASK TYPE
Classification
Generate Recommendations
150 ROWS • 4 FEATURES • TARGET_NAME TARGET
Top Recommendations
Gradient Boosting
E[Win Rate]: 81.0%
Confidence: 36.0%
Strong on tabular data, good bias-variance trade-off
Train this model
Random Forest
E[Win Rate]: 80.0%
Confidence: 33.4%
Robust ensemble, handles mixed features well, rarely overfits
Lightgbm
E[Win Rate]: 72.0%
Confidence: 10.4%
Fastest on large datasets, native categorical support
4th Xgboost
E[Win Rate]: 72.0%
Confidence: 9.4%
State-of-the-art on structured data, handles missing values
5th Svm
E[Win Rate]: 65.0%
Confidence: 5.4%
Effective in high-dimensional spaces with clear margins
Bandit Posterior
0 RECORDED OUTCOMES
MODEL POSTERIOR MEAN WIN RATE A B
gradient boosting 81.0% 8.5 2.0
random forest 80.0% 8.0 2.0
xgboost 72.0% 9.0 3.5
lightgbm 72.0% 9.0 3.5
logistic regression 65.2% 7.5 4.0
svm 65.0% 6.5 3.5
knn 54.5% 6.0 5.0
decision tree 50.0% 6.0 6.0
naive bayes 45.0% 4.5 5.5
HOW IT WORKS
Each model maintains a Beta(α, β) prior distribution. α increases on successful training outcomes (high metric), while β increases on poor outcomes. The mean α/(α+β) represents the estimated win rate. Thompson Sampling probabilistically selects models by drawing from these distributions—perfectly balancing exploration of new models with exploitation of known good ones.

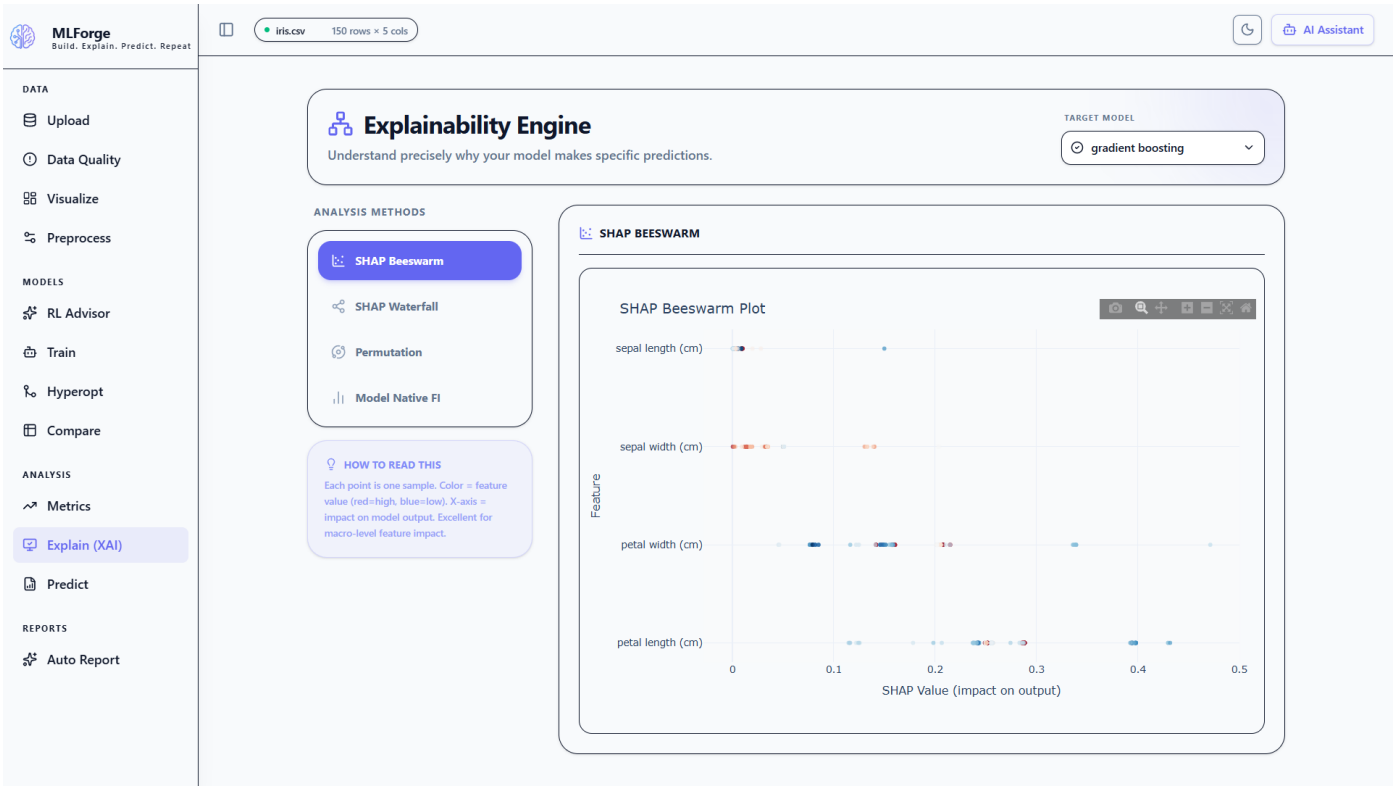

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Model Training
4 features → target_name • CLASSIFICATION
SELECT ALGORITHM
Logistic Regression
CLASSIFICATION
Random Forest
CLASSIFICATION • REGRESSION
Gradient Boosting
CLASSIFICATION • REGRESSION
XGBoost
CLASSIFICATION • REGRESSION
LightGBM
CLASSIFICATION • REGRESSION
Support Vector Machine
Spawn Training Job
TRAINING RUNS TRACKER
2 RUNS
Gradient Boosting
COMPLETE
ACCURACY 96.7%
4.1S
VIEW EVALUATION METRICS →
Xgboost
COMPLETE
ACCURACY 93.3%
20.7S
VIEW EVALUATION METRICS →
xgboost training complete!

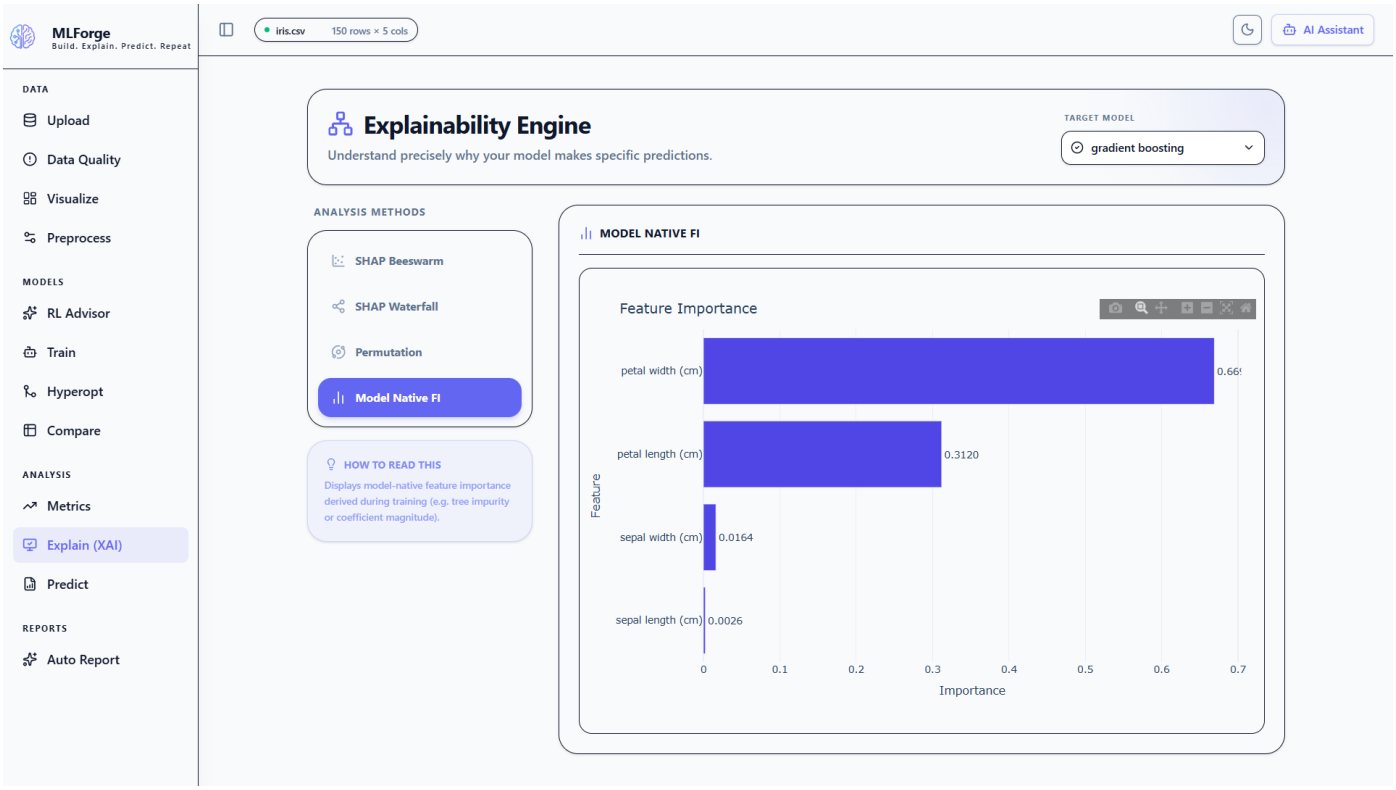

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Hyperparameter Tuning
Optuna TPE search — 4 features → target_name (classification)
Configure Search
Model
random forest
Trials: 30
10
100
Timeout: 120s
Start Search
Search
Configure and start a hyperparameter search to find the best settings for your model.

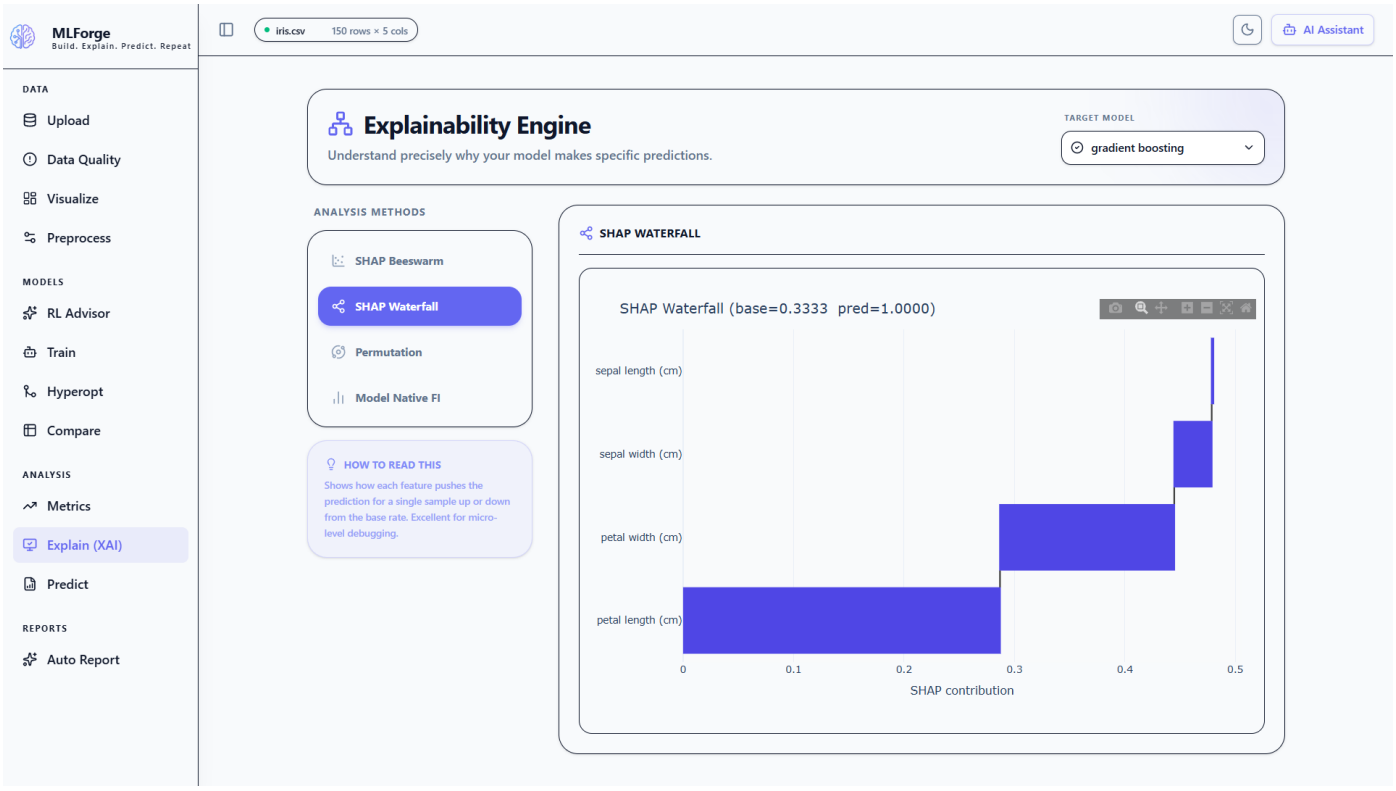

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Model Leaderboard
2 completed runs ready for comparison
Compare All Models
RANK
MODEL ARCHITECTURE
ACCURACY
F1 WEIGHTED
PRECISION WEIGHTED
RECALL WEIGHTED
AUC ROC
DURATION
DETAILS
Gradient Boosting
96.67%
96.7%
97.0%
96.7%
99.0%
4.12S
METRICS
Xgboost
93.33%
93.3%
93.3%
93.3%
96.5%
20.66S
METRICS
Leaderboard updated

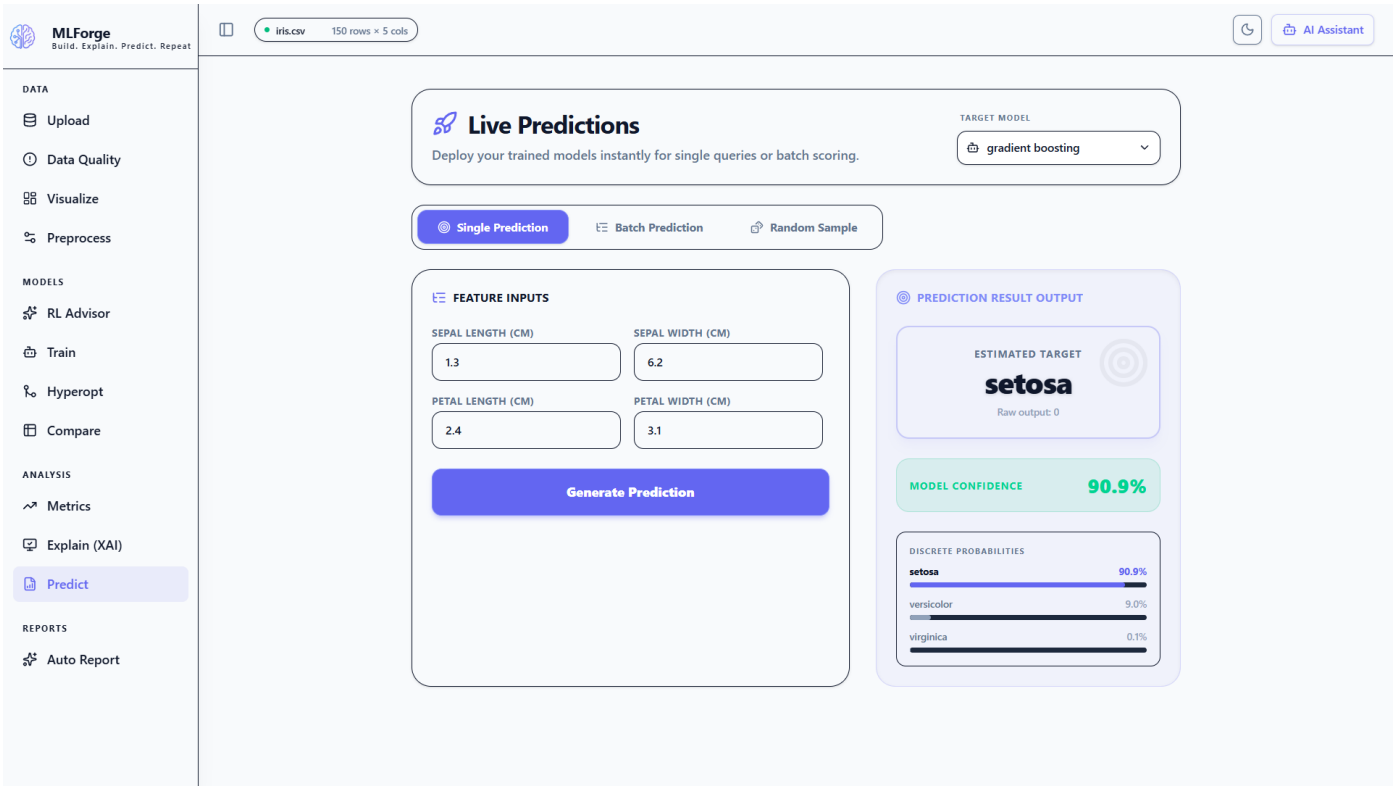

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Evaluation Dashboard
Gradient Boosting • CLASSIFICATION • TARGET_NAME
gradient boosting
Export .pkl
Live Predict
96.7% ACCURACY
96.7% F1 WEIGHTED
96.7% F1 MACRO
97.0% PRECISION WEIGHTED
96.7% RECALL WEIGHTED
95.2% MCC
99.0% AUC ROC
36.3% LOG LOSS
CONFUSION MATRIX
Confusion Matrix
Actual
Predicted
setosa
versicolor
virginica
ROC CURVE
ROC Curve (One-vs-Rest)
True Positive Rate
False Positive Rate

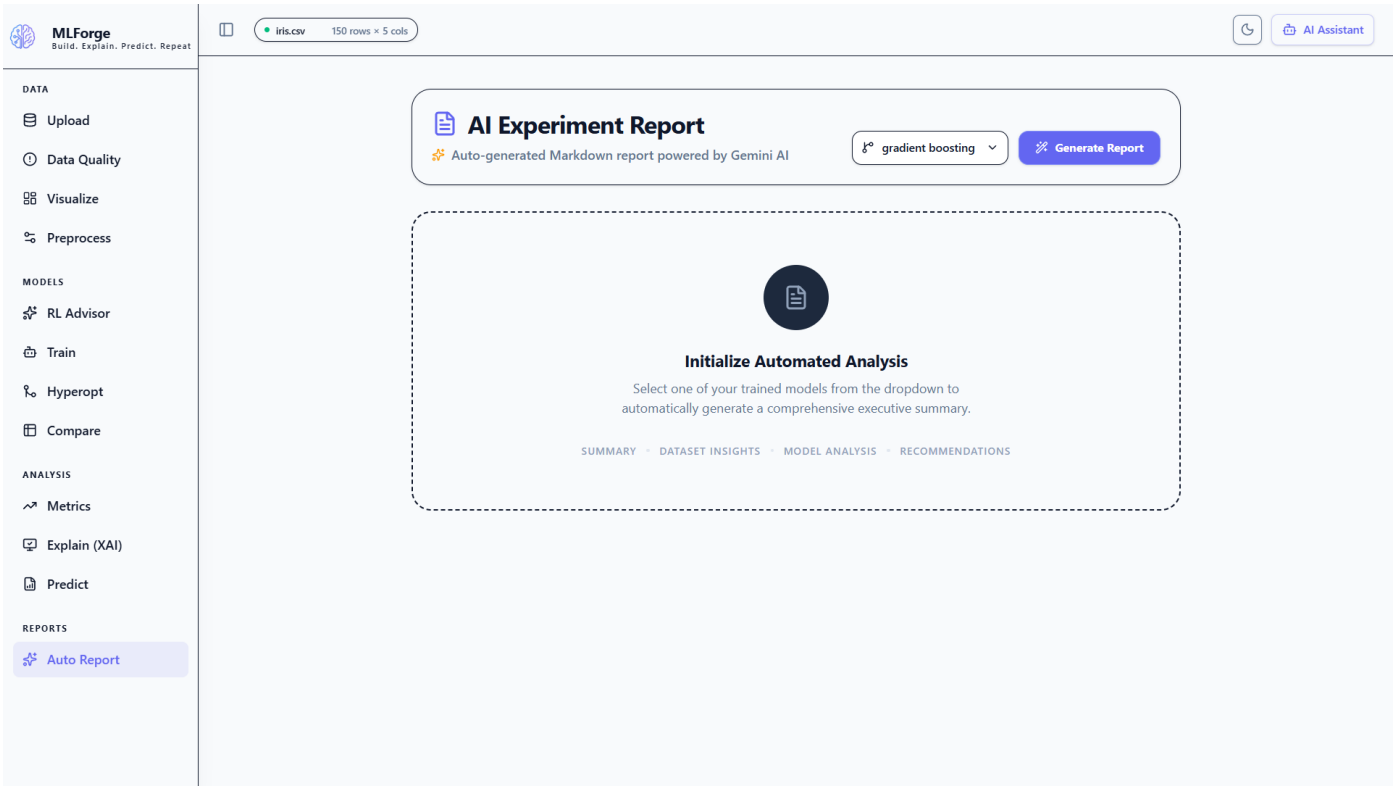

MLForge
Build. Explain. Predict. Repeat
iris.csv 150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
FEATURE IMPORTANCE
Feature Importance
petal width (cm) 0.6691
petal length (cm) 0.3120
sepal width (cm) 0.0164
sepal length (cm) 0.0026
Feature
Importance
CROSS-VALIDATION SCORES
FOLD 1 95.8%
FOLD 2 95.8%
FOLD 3 95.8%
FOLD 4 95.8%
FOLD 5 95.8%
VALIDATION VARIANCE
Mean: 95.83% ± 0.00%

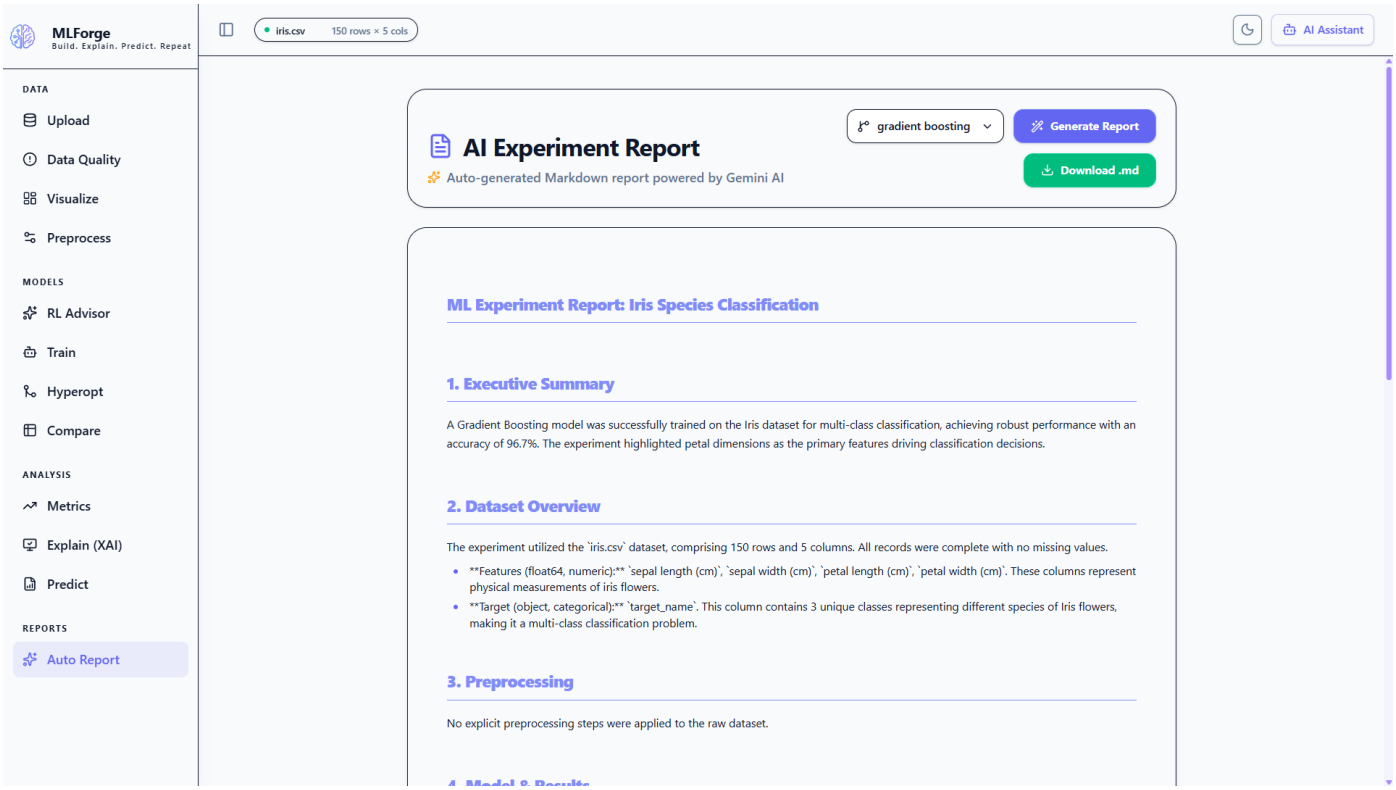

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Explainability Engine
Understand precisely why your model makes specific predictions.
TARGET MODEL
gradient boosting
ANALYSIS METHODS
SHAP Beeswarm
SHAP Waterfall
Permutation
Model Native FI
HOW TO READ THIS
Each point is one sample. Color = feature value (red=high, blue=low). X-axis = impact on model output. Excellent for macro-level feature impact.
SHAP BEESWARM
SHAP Beeswarm Plot
sepal length (cm)
sepal width (cm)
petal width (cm)
petal length (cm)
Feature
0
0.1
0.2
0.3
0.4
0.5
SHAP Value (impact on output)

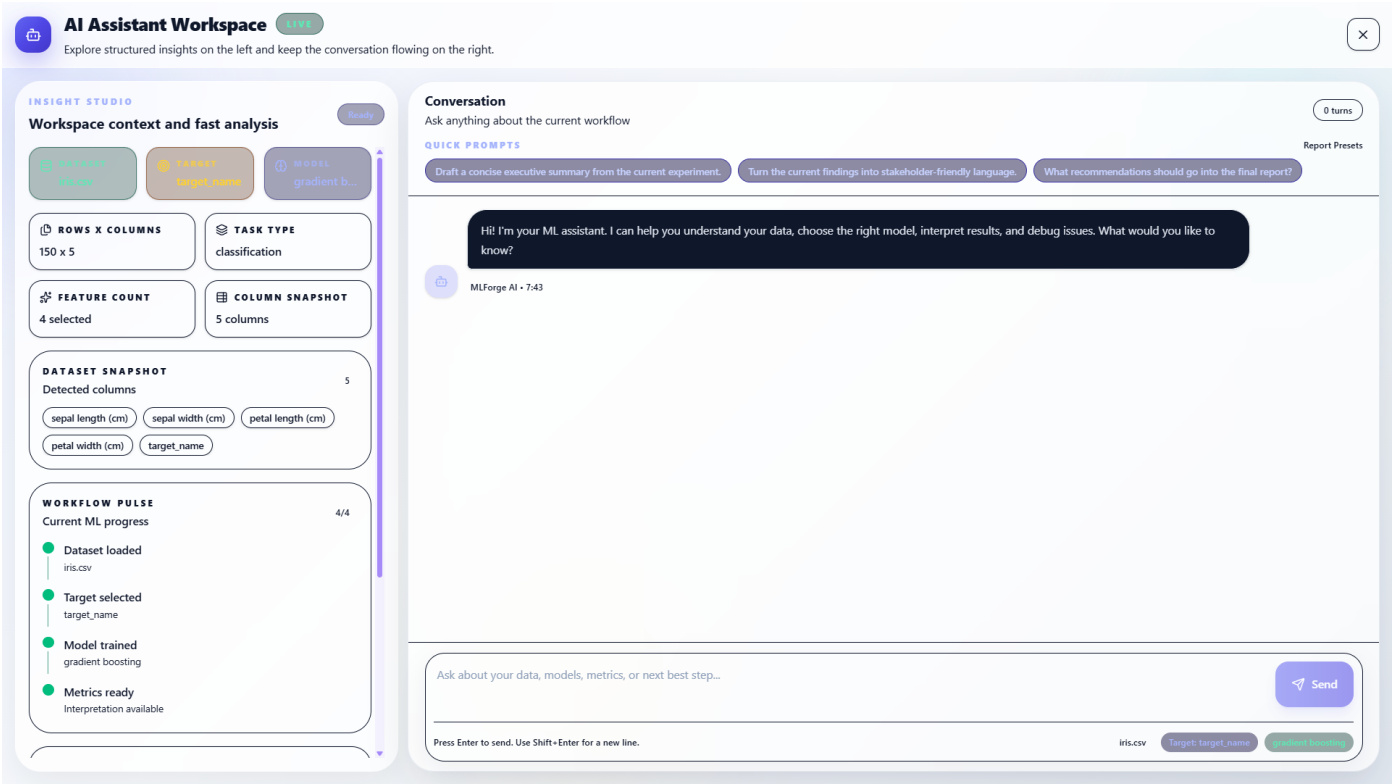

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Explainability Engine
Understand precisely why your model makes specific predictions.
TARGET MODEL
gradient boosting
ANALYSIS METHODS
SHAP Beeswarm
SHAP Waterfall
Permutation
Model Native FI
HOW TO READ THIS
Displays model-native feature importance derived during training (e.g. tree impurity or coefficient magnitude).
MODEL NATIVE FI
Feature Importance
petal width (cm)
0.66
petal length (cm)
0.3120
sepal width (cm)
0.0164
sepal length (cm)
0.0026
Feature
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Importance

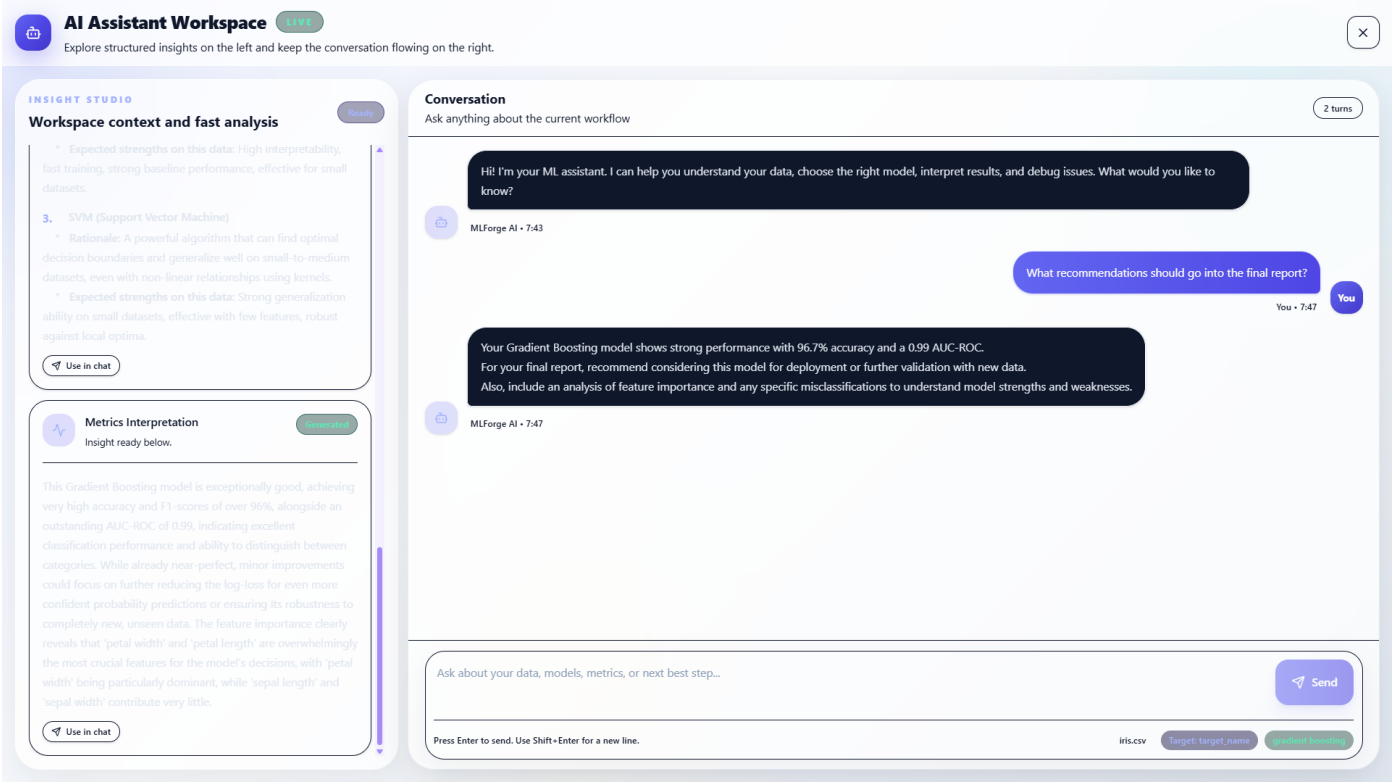

MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Explainability Engine
Understand precisely why your model makes specific predictions.
TARGET MODEL
gradient boosting
ANALYSIS METHODS
SHAP Beeswarm
SHAP Waterfall
Permutation
Model Native FI
HOW TO READ THIS
Shows how each feature pushes the prediction for a single sample up or down from the base rate. Excellent for micro-level debugging.
SHAP WATERFALL
SHAP Waterfall (base=0.3333 pred=1.0000)
sepal length (cm)
sepal width (cm)
petal width (cm)
petal length (cm)
0
0.1
0.2
0.3
0.4
0.5
SHAP contribution


MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
Live Predictions
Deploy your trained models instantly for single queries or batch scoring.
TARGET MODEL
gradient boosting
Single Prediction
Batch Prediction
Random Sample
FEATURE INPUTS
SEPAL LENGTH (CM)
1.3
SEPAL WIDTH (CM)
6.2
PETAL LENGTH (CM)
2.4
PETAL WIDTH (CM)
3.1
Generate Prediction
PREDICTION RESULT OUTPUT
ESTIMATED TARGET
setosa
Raw output: 0
MODEL CONFIDENCE
90.9%
DISCRETE PROBABILITIES
setosa
90.9%
versicolor
9.0%
virginica
0.1%


MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
AI Experiment Report
Auto-generated Markdown report powered by Gemini AI
gradient boosting
Generate Report
Initialize Automated Analysis
Select one of your trained models from the dropdown to automatically generate a comprehensive executive summary.
SUMMARY · DATASET INSIGHTS · MODEL ANALYSIS · RECOMMENDATIONS


MLForge
Build. Explain. Predict. Repeat
iris.csv
150 rows × 5 cols
AI Assistant
DATA
Upload
Data Quality
Visualize
Preprocess
MODELS
RL Advisor
Train
Hyperopt
Compare
ANALYSIS
Metrics
Explain (XAI)
Predict
REPORTS
Auto Report
AI Experiment Report
Auto-generated Markdown report powered by Gemini AI
gradient boosting
Generate Report
Download .md
ML Experiment Report: Iris Species Classification
1. Executive Summary
A Gradient Boosting model was successfully trained on the Iris dataset for multi-class classification, achieving robust performance with an accuracy of 96.7%. The experiment highlighted petal dimensions as the primary features driving classification decisions.
2. Dataset Overview
The experiment utilized the `iris.csv` dataset, comprising 150 rows and 5 columns. All records were complete with no missing values.
• **Features (float64, numeric):** `sepal length (cm)`, `sepal width (cm)`, `petal length (cm)`, `petal width (cm)`. These columns represent physical measurements of iris flowers.
• **Target (object, categorical):** `target_name`. This column contains 3 unique classes representing different species of Iris flowers, making it a multi-class classification problem.
3. Preprocessing
No explicit preprocessing steps were applied to the raw dataset.


AI Assistant Workspace
LIVE
Explore structured insights on the left and keep the conversation flowing on the right.
INSIGHT STUDIO
Workspace context and fast analysis
Ready
DATASET
iris.csv
TARGET
target_name
MODEL
gradient b...
ROWS X COLUMNS
150 x 5
TASK TYPE
classification
FEATURE COUNT
4 selected
COLUMN SNAPSHOT
5 columns
DATASET SNAPSHOT
Detected columns
5
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)
target_name
WORKFLOW PULSE
Current ML progress
4/4
Dataset loaded
iris.csv
Target selected
target_name
Model trained
gradient boosting
Metrics ready
Interpretation available
Conversation
Ask anything about the current workflow
0 turns
QUICK PROMPTS
Report Presets
Draft a concise executive summary from the current experiment.
Turn the current findings into stakeholder-friendly language.
What recommendations should go into the final report?
Hi! I'm your ML assistant. I can help you understand your data, choose the right model, interpret results, and debug issues. What would you like to know?
MLForge AI • 7:43
Ask about your data, models, metrics, or next best step...
Send
Press Enter to send. Use Shift+Enter for a new line.
iris.csv
Target: target_name
gradient boosting


AI Assistant Workspace
LIVE
Explore structured insights on the left and keep the conversation flowing on the right.
INSIGHT STUDIO
Workspace context and fast analysis
Ready
* Expected strengths on this data: High interpretability, fast training, strong baseline performance, effective for small datasets.
3. SVM (Support Vector Machine)
* Rationale: A powerful algorithm that can find optimal decision boundaries and generalize well on small-to-medium datasets, even with non-linear relationships using kernels.
* Expected strengths on this data: Strong generalization ability on small datasets, effective with few features, robust against local optima.
Use in chat
Metrics Interpretation
Insight ready below.
Generated
This Gradient Boosting model is exceptionally good, achieving very high accuracy and F1-scores of over 96%, alongside an outstanding AUC-ROC of 0.99, indicating excellent classification performance and ability to distinguish between categories. While already near-perfect, minor improvements could focus on further reducing the log-loss for even more confident probability predictions or ensuring its robustness to completely new, unseen data. The feature importance clearly reveals that 'petal width' and 'petal length' are overwhelmingly the most crucial features for the model's decisions, with 'petal width' being particularly dominant, while 'sepal length' and 'sepal width' contribute very little.
Use in chat
Conversation
Ask anything about the current workflow
2 turns
Hi! I'm your ML assistant. I can help you understand your data, choose the right model, interpret results, and debug issues. What would you like to know?
MLForge AI • 7:43
What recommendations should go into the final report?
You
You • 7:47
Your Gradient Boosting model shows strong performance with 96.7% accuracy and a 0.99 AUC-ROC.
For your final report, recommend considering this model for deployment or further validation with new data.
Also, include an analysis of feature importance and any specific misclassifications to understand model strengths and weaknesses.
MLForge AI • 7:47
Ask about your data, models, metrics, or next best step...
Send
Press Enter to send. Use Shift+Enter for a new line.
iris.csv
Target: target_name
gradient boosting

CHAPTER 5: Results & Conclusion

5.1. Result

Predictions with high accuracy:

The implemented system efficiently trains and evaluates multiple machine learning models, leading to reliable predictions. For example, models such as Random Forest and Logistic Regression show high accuracy on test datasets, making the system reliable for real-world applications.

Extensive data analysis:

Users can perform end-to-end analysis of datasets, including cleaning data, handling missing values, detecting outliers, and generating descriptive statistics. The platform simplifies tasks that typically require multiple tools and technical expertise.

Interactive visualization:

The system provides dynamic visualizations such as bar charts, line plots, dot plots, histograms and pie charts. Users can filter, sort and drill down on data to quickly identify trends, correlations and insights.

Model assessment:

Multiple system getting to know models can be trained simultaneously, and overall performance metrics along with accuracy, precision, keep in mind, F1 rating, and ROC-AUC are displayed for smooth contrast. It helps in selecting the most suitable version for a given records set.

ease of use:

The GUI ensures an intuitive experience, allowing even non-technical users to explore data, train models and generate predictions without writing any code.

5.2. Benefits

Time efficiency:

Automates repetitive tasks including information preprocessing, visualization and version assessment, saving large time for users.

Consumer-pleasant interface:

Streamlit's interactive interface presents clean navigation and short responses, lowering the gaining knowledge of curve for information technological know-how beginners.

Resilience:

Supports a couple of dataset codecs (CSV, Excel, TXT) and diverse ML algorithms, making it well ideal for loads of analysis and prediction issues.

Scalable Architecture:

The modular layout permits new features, algorithms and visualizations to be delivered in destiny iterations with out massive redesign.

Availability:

Since it is web-based, the platform can be accessed from anywhere, on any device, without complicated installation or configuration.

Data-driven decision making:

By providing actionable insights through visualization and predictions, the system empowers organizations and individuals to make informed decisions based on data rather than intuition.

5.3. Future Improvements

Integration of advanced ML and AI models:

Include deep mastering fashions including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN) or Recurrent Neural Networks (RNN) to address more complicated datasets inclusive of photos, text or time series information.

Real-time information evaluation and forecasting:

Implement sreaming facts support to offer immediately analysis and forecasting, which is beneficial for applications along with inventory marketplace analysis, IoT tracking, and actual-time reporting.

Customizable Dashboard:

Allow customers to layout their own dashboards with favorite charts, metrics and reviews, and keep non-public workflows for repeated evaluation.

Advanced consumer assist:

Add step-by means of-step publications, tooltips, and AI-powered characteristic pointers to assist customers understand and use the platform greater effectively.

Collaboration features:

Shared datasets enable a couple of customers to paintings collectively on the same challenge, with model manipulate and collaborative insights, enhancing teamwork for companies.

Cloud deployment and integration:

Deploy the device on a cloud platform along with AWS, Google Cloud or Azure to boom computing strength, garage and attain, making it suitable for massive-scale corporation use.

Automated reporting:

Generate automated reports in PDF or Excel with visualizations, model performance metrics and predictions for easy sharing and documentation.

Security and privacy improvements:

Implement authentication, data encryption and secure storage protocols to ensure the security of sensitive data during analysis and storage.

REFERENCES

- Research Papers & Journals
- Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques. Elsevier.
- Witten, I., Frank, E., & Hall, M. (2016). Data Mining: Practical Machine Learning Tools and Techniques. Morgan Kaufmann.
- Breiman, L. (2001). Random Forests. Machine Learning Journal.
- Domingos, P. (2012). A Few Useful Things to Know About Machine Learning. Communications of the ACM.
- Zhang, Y. et al. (2020). Interactive Machine Learning for Data Visualization. IEEE Transactions on Visualization.
- Kuhn, M., & Johnson, K. (2013). Applied Predictive Modeling. Springer.
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD Conference.
- Fayyad, U. et al. (1996). From Data Mining to Knowledge Discovery. AI Magazine.
- Jordan, M., & Mitchell, T. (2015). Machine Learning Trends. Science Journal.
- Zhang, D. (2019). Data Preprocessing Techniques in ML. Journal of Intelligent Systems.

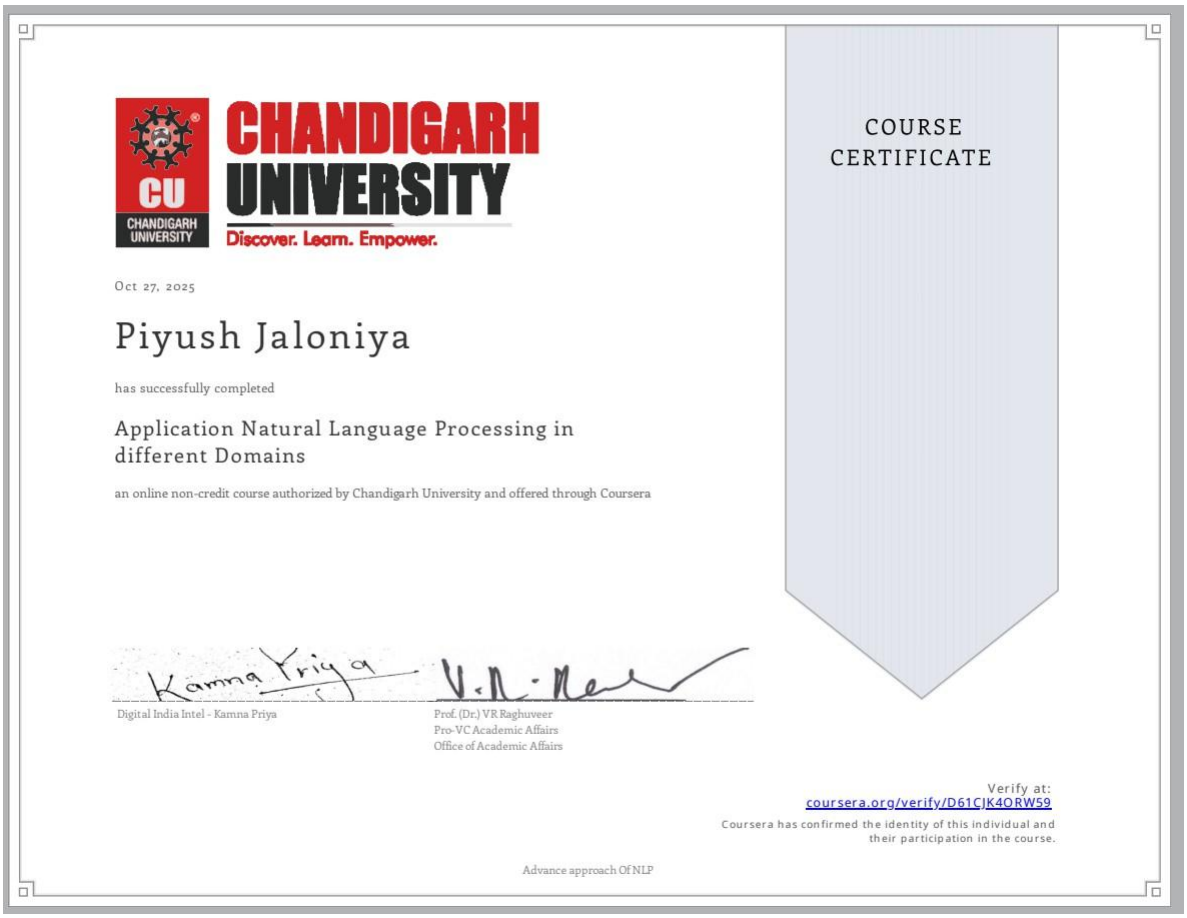
Books & Learning Sources

- Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn and TensorFlow. O'Reilly.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
- Raschka, S. (2015). Python Machine Learning. Packt Publishing.
- Chollet, F. (2018). Deep Learning with Python. Manning Publications.
- Banday, S. (2021). Introduction to Machine Learning. Pearson.

Websites & Online Resources

- Scikit-learn Documentation – https://scikit-learn.org
- Kaggle ML Guides – https://www.kaggle.com/learn
- Towards Data Science Articles – https://towardsdatascience.com
- Analytics Vidhya ML Tutorials – https://www.analyticsvidhya.com
- TensorFlow Documentation – https://www.tensorflow.org

Coursera Certificates



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Analysis Report

Plagiarism Detection and AI Detection Report

MLForge-Project_Report.pdf

Plagiarism Detection

0%

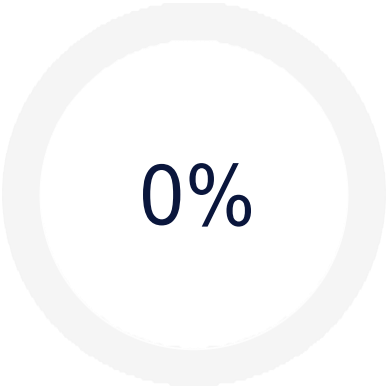
Plagiarism Types	Text Coverage	Words
Identical	0%	0
Minor Changes	0%	0
Paraphrased	0%	0
Excluded		
Omitted Words		85

AI Detection

16%

	Text Coverage	Words
AI Text	16%	2,055
Low Frequency		413
Medium Frequency		0
High Frequency		3
Human Text	84%	10,824
Excluded		
Omitted Words		85

Plagiarism



Results (0)

*Results may not appear because the feature has been disabled.

Private Cloud Hub	Shared Data Hub	Filtered / Excluded
0	0	0
Internet Sources	**AI Source Match**	**Current Batch**
0	0	0

Plagiarism Types	Text Coverage	Words
Identical	0%	0
Minor Changes	0%	0
Paraphrased	0%	0
Excluded		
Omitted Words		85

About Plagiarism Detection

Our AI-powered plagiarism scans offer three layers of text similarity detection: Identical,Minor Changes,and Paraphrased.Based on your scan settings we also provide insight on how much of the text you are not scanning for plagiarism (Omitted words).

Identical

One to one exact word matches. Learn more

Minor Changes

Words that hold nearly the same meaning but have a change to their form (e.g."large" becomes"largely"). Learn more

Paraphrased

Different words that hold the same meaning that replace the original content (e.g. "large" becomes"big") Learn more

Omitted Words

The portion of text that is not being scanned for plagiarism based on the scan settings. (e.g.the'Ignore quotations'setting is enabled and the document is 20% quotations making the omitted words percentage 20%) Learn more

Copyleaks Shared Data Hub

Our Shared Data Hub is a collection of millions of user-submitted documents that you can utilize as a scan resource and choose whether or not you would like to submit the file you are scanning into the Shared Data Hub. Learn more

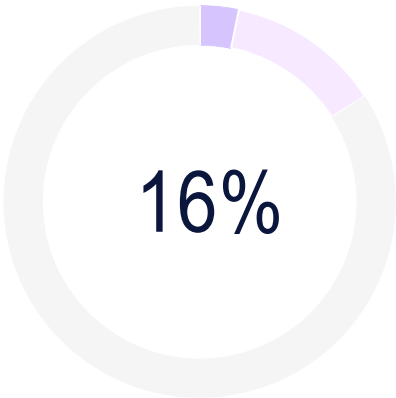
Filtered and Excluded Results

The report will generate a complete list of results.There is always the option to exclude specific results that are not relevant.Note,by unchecking certain results,the similarity percentage may change. Learn more

Current Batch Results

These are the results displayed from the collection,or batch,of files uploaded for a scan at the same time. Learn more

AI Content



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Excluded		
Omitted Words		85

About AI Detection

Our AI Detector is the only enterprise-level solution that can verify if the content was written by a human or generated by AI,including source code and text that has been plagiarized or modified. Learn more

AI Text

A body of text that has been generated or altered by AI technology. Learn more

Human Text

Any text that has been fully written by a human and has not been altered or generated by AI. Learn more

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Copyleaks AI Detector Effectiveness

Credible data at scale,coupled with machine learning and widespread adoption,allows us to continually refine and improve our ability to understand complex text patterns, resulting in over 99% accuracy—far higher than any other AI detector—and improving daily. Learn more

Ideal Text Length

The higher the character count,the easier for our technology to determine irregular patterns,which results in a higher confidence rating for AI detection. Learn more

Reasons It Might Be AI When You Think It's Not

The AI Detector can detect a variety of AI-generated text,including tools that use AI technology to paraphrase content,auto-complete sentences,and more. Learn more

User AI Alert History

Historical data of how many times a user has been flagged for potentially having AI text within their content. Learn more

AI Logic

The number of times a phrase was found more frequently in AI vs human text is shown according to low,medium,and high frequency. Learn more